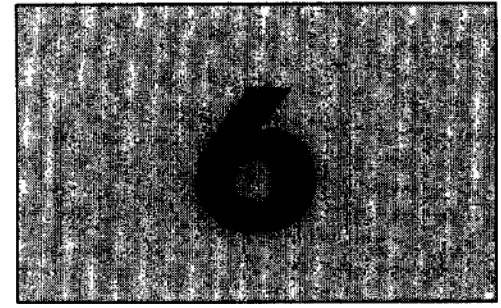


Order and broken symmetry



The appearance of spontaneous order at low temperature is a fundamental phenomenon of condensed matter physics. Ferromagnets, antiferromagnets, liquid crystals and superconductors are all ordered phases, as is the solid state itself. All these phenomena share some fundamental properties and characteristics. For example, they are all characterized by a temperature dependence in which some relevant physical property shows a marked difference above and below a critical temperature T_c . For each phase one can define an order parameter which is zero for $T > T_c$ and non-zero for $T < T_c$. This quantity therefore acts as an indicator of whether or not the system is ordered. In the case of ferromagnetism, the order parameter is simply the magnetization. In this chapter we will show that each type of ordered phase is associated with a broken symmetry, a concept which is explained in the following section.

6.1 Broken symmetry

In ferromagnets a single unique direction has been chosen along which all the atomic magnetic moments have lined up. They have all chosen 'up' rather than 'down'. This is actually a rather surprising effect since the underlying physical equations do not distinguish between 'up' and 'down'. Thus the microscopic physics has a symmetry not possessed by the experimentally observed ground state. To delve further into this mystery, we must consider symmetry in greater detail.

A useful example to consider is the Euler strut, shown in Fig. 6.1. This consists of a vertical rod which is clamped into the ground at the bottom and uniformly loaded from the top. As the weight of the load increases, the rod compresses, and above a critical weight it buckles. Whether it buckles to the left or the right depends on the precise details of exactly how symmetrically one places the load on the top. But even in the idealization that it is placed perfectly centrally on the top, the rod still has to choose which way to buckle. Balanced precariously, as if on a knife-edge, a random thermal fluctuation might be enough to send it buckling one way or the other. The left-right symmetry, inherent in the lightly loaded rod, is said to be **broken** by the buckling.

Many similar features are observed in condensed matter systems. The parameter which drives the symmetry-breaking transition can be a force, such as an applied pressure, but it is very often temperature. Figure 6.2 shows atoms in a liquid and in a solid. As a liquid cools there is a very slight contraction of the system but it retains a very high degree of symmetry. However, below a critical temperature, the melting temperature, the liquid becomes a solid and that symmetry is broken. This may at first sight seem surprising because the picture

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Terminology concerning symmetry

A system possesses a particular symmetry if the Hamiltonian describing it is invariant with respect to the transformations associated with elements of a **symmetry group**. A group is a collection of elements and an operation which combines them, that follow a particular set of rules: the group is 'closed', so that combinations of elements are also members of the group; the group combination is associative; an inverse and identity are well defined.

A **discrete symmetry** refers to a symmetry group with countable elements, such as the rotational symmetry group of a cube. A **continuous symmetry** has an uncountable continuum of elements, such as the rotational symmetry group of a sphere. A system possesses a **global symmetry** if it is invariant under the symmetry elements of the group being applied globally to the entire system. A **local symmetry** applies to the Hamiltonian if it is unchanged after the symmetry operations are applied differently to different points in space. The gauge symmetry of superconductivity is such a local symmetry.

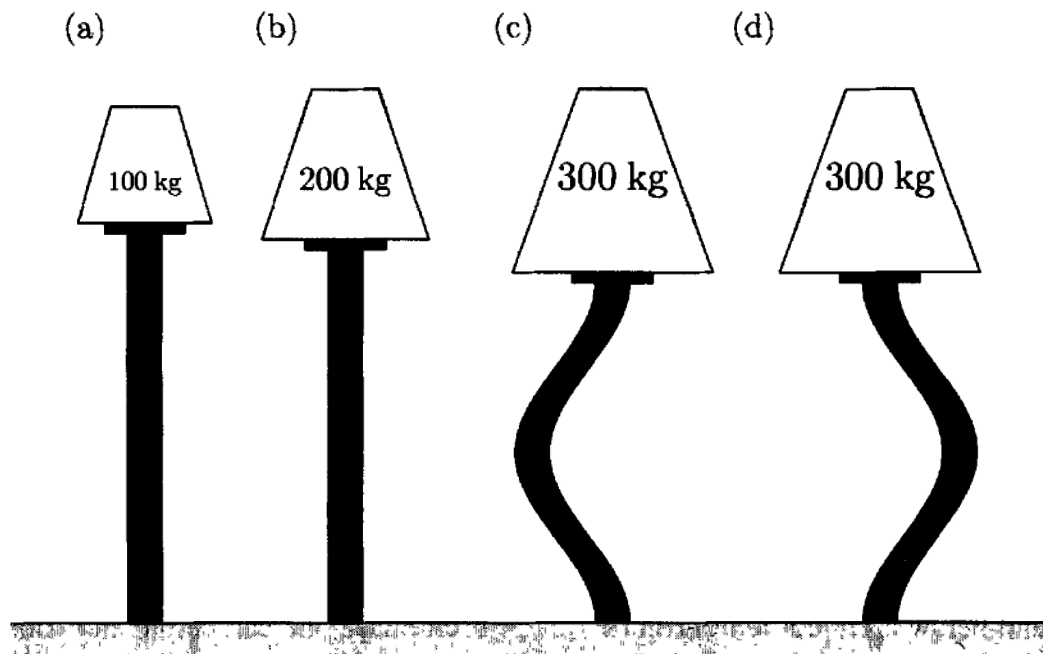


Fig. 6.1 (a) The Euler strut. (b) As an increasing vertical force is applied to the strut, the strut is compressed but retains left-right symmetry. (c) Once the force exceeds a critical value, the strut buckles, breaking the left-right symmetry. (d) The strut could also break the symmetry in the opposite sense.

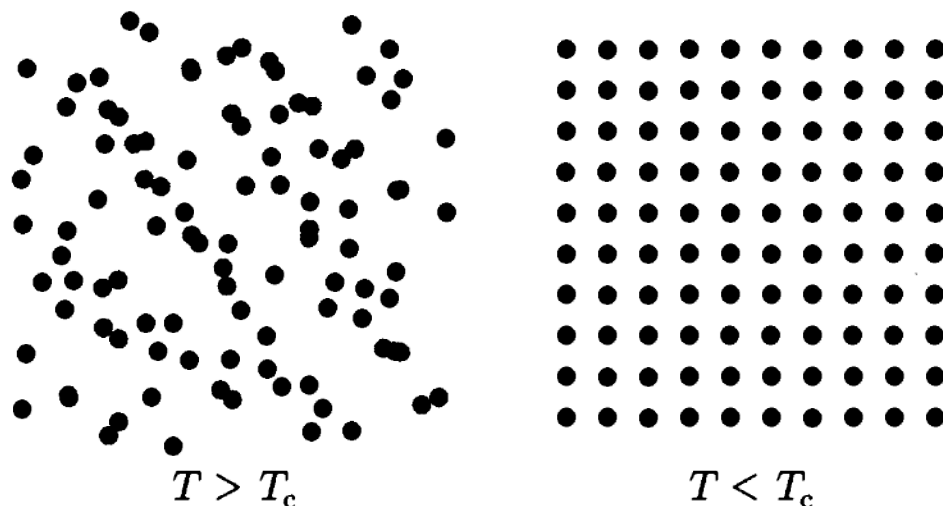


Fig. 6.2 Liquid-solid phase transition. Left: The high temperature state (statistically averaged) has complete translational and rotational symmetry. Right: These symmetries are broken as the system becomes a solid below the critical temperature T_c .

of the solid ‘looks’ more symmetrical than that of the liquid. The atoms in the solid are all symmetrically lined up while in the liquid they are all over the place. The crucial observation is that any point in a liquid is exactly the same as any other. If you average the system over time, each position is visited by atoms as often as any other. There are no unique directions or axes along which atoms line up. In short, the system possesses complete translational and rotational symmetry. In the solid however this high degree of symmetry is nearly all lost. The solid drawn in Fig. 6.2 still possesses some residual symmetry: rather than being invariant under arbitrary rotations, it is invariant under four-fold rotations ($\pi/2, \pi, 3\pi/2, 2\pi$); rather than being invariant under arbitrary translations, it is now invariant under a translation of an integer combination of lattice basis vectors. Therefore not all symmetry has been lost but the high symmetry of the liquid state has been broken.

The situation is similar for a ferromagnet (Fig. 6.3). A ferromagnet above the Curie temperature T_C possesses complete rotational symmetry. All directions are equivalent and each magnetic moment can point in any direction. Below T_C the system ‘chooses’ a unique direction for all the spins to point. The

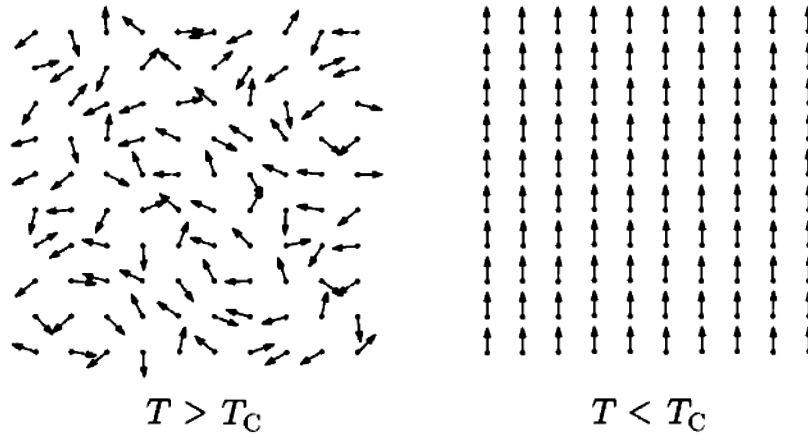


Fig. 6.3 Paramagnet–ferromagnet phase transition. Left: The high temperature state (statistically averaged) has rotational symmetry. Right: This symmetry is broken as the system becomes a ferromagnet below the Curie temperature T_C .

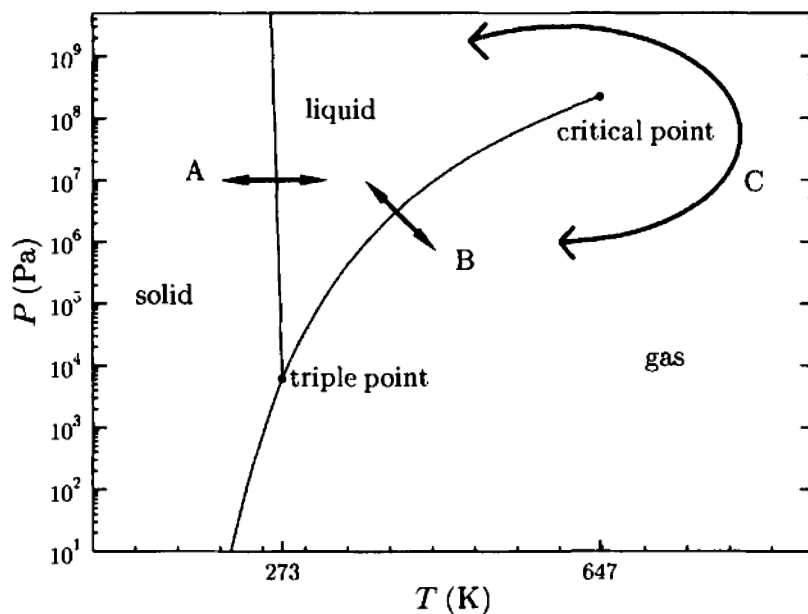


Fig. 6.4 The phase diagram of water. Paths A and B in the phase diagram cross a phase boundary, while C does not. Path A involves a change of symmetry, while B and C do not.

higher rotational symmetry of the high temperature state is broken. The lower temperature state has a reduced rotational symmetry (only rotations about the magnetization axis, the up-direction in Fig. 6.3, are allowed).

An important point to notice is that it is impossible to change symmetry *gradually*. Either a particular symmetry is present or it is not. Hence, phase transitions are sharp and there is a clear delineation between the ordered and disordered states. The appearance of order at low temperatures can then be understood from quite general thermodynamic considerations. The free energy F is related to the energy E and the entropy S by $F = E - TS$. In order to minimize the free energy, at low temperature a system will choose its lowest energy ground state which is usually ordered. Thus it will minimize E . As the temperature increases, it becomes more important in minimizing F to find a state which maximizes S and so a disordered state is favoured.

Not all phase transitions involve a change of symmetry. Figure 6.4 shows the phase diagram of water. The boundary line between the liquid and gas regions is terminated by a **critical point**. Hence it is possible to ‘cheat’ the

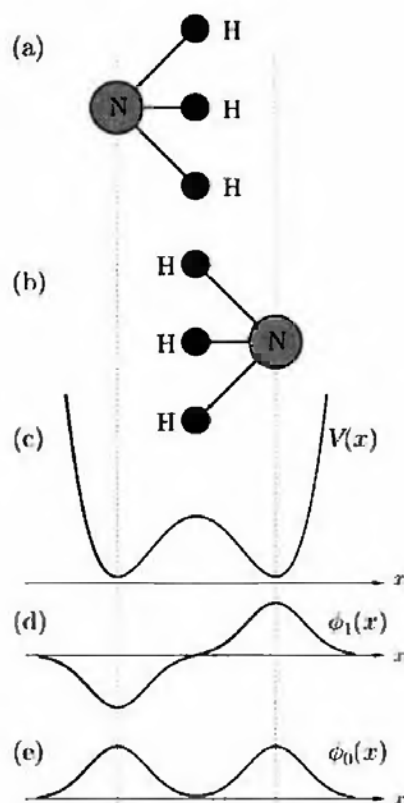


Fig. 6.5 (a) Ammonia, NH_3 , is a tetrahedral molecule which can exist in one of two forms. The other form is shown in (b). There are two stable positions of the N atom, and we can thus define a potential energy $V(x)$ associated with this system. (d) The first excited state wave function $\phi_1(x)$. (e) The ground state wave function $\phi_0(x)$.

sharp phase transition by taking a path through the phase diagram which avoids a discontinuous change (path C in Fig. 6.4). For temperatures above the critical temperature (647 K) a gas and a liquid are distinguished only by their density. The transition between a gas and a liquid involves no change of symmetry. In contrast, the solid-liquid transition involves a change of symmetry and consequently there is no critical point for the melting curve.

A puzzle about the nature of symmetry-breaking transitions is that the broken-symmetry ground state does not possess the symmetry of the Hamiltonian. On cooling a ferromagnet through the transition temperature, the system has to choose a particular direction along which all the spins will point, even though no particular direction is singled out in the underlying physics. How can this come about?

To understand this, it is useful to consider a non-magnetic example, that of the molecule ammonia (NH_3) shown in Fig. 6.5. The three hydrogen atoms form three corners of an equilateral triangle (viewed edge-on in Figures 6.5(a) and 6.5(b)) and the nitrogen atom can be just above the plane of the triangle or just below it. This pyramidal shape is due to the lone pair of electrons on the nitrogen which jockeys for space with the three nitrogen-hydrogen bonds. This gives the molecule a dipole moment, but also defines a particular direction in space and breaks the symmetry of the underlying Hamiltonian. Therefore this pyramid cannot be a stable state of the system.

In fact we can imagine transforming between two pyramidal states shown in Figures 6.5(a) and (b) by moving the nitrogen through the middle of the plane, thus inverting the tetrahedron much in the same way as an umbrella is inverted when it is blown out by a strong wind. If we denote the position of the nitrogen atom by x , the distance from the plane, we could imagine a potential energy curve $V(x)$ which has two minima associated with the two stable configurations. This is shown in Fig. 6.5(c). The two lowest energy eigenstates of this potential are shown in Figures 6.5(d) and (e) for the first excited state $\phi_1(x)$ and ground state $\phi_0(x)$ respectively.

The configurations represented in Fig. 6.5(a) and (b) do not therefore represent stable configurations of the system, but are superpositions of energy eigenstates. They therefore represent *metastable states*. Thus the state in Fig. 6.5(a) is represented by $(\phi_0(x) - \phi_1(x))/\sqrt{2}$ and the state in Fig. 6.5(b) is represented by $(\phi_0(x) + \phi_1(x))/\sqrt{2}$. If you prepared the system in the metastable state in Fig. 6.5(a), you would find that it would oscillate between that and the metastable state in Fig. 6.5(b) at a frequency given by the difference between the energy of $\phi_0(x)$ and $\phi_1(x)$. This frequency is 24 GHz in ammonia and this transition is the one used in the ammonia maser. If you replace the nitrogen for the more massive phosphorous to make PH_3 , the frequency goes down by about a factor of ten. Thus as we make the system heavier we *increase* the time constant of the metastable state in Fig. 6.5(a).

In a ferromagnet we have a more massive state yet, involving not just heavier atoms, but more atoms, maybe 10^{23} of them. The ground state of the system with all spins pointing in a single direction is actually not a true stationary state of the system. However it is a metastable state but with a stability time which is greater than the age of the universe. To access the other state, the one with all spins reversed, one would have to flip every spin in the system simultaneously, which is extremely unlikely.

6.2 Models

In order to discuss some of the consequences of symmetry breaking it is convenient to think about some simple *models* of magnetism and try to solve them.

6.2.1 Landau theory of ferromagnetism

A convenient model which simply produces a phase transition was provided by the Russian physicist Lev Landau and arises from some very general considerations. We write down the free energy for a ferromagnet with magnetization M as a power series in M . Because there is no energetic difference between 'up' or 'down', this power series cannot contain any odd power of M . Therefore we can write for the free energy $F(M)$ the expression

$$F(M) = F_0 + a(T)M^2 + bM^4 \quad (6.1)$$

where F_0 and b are constants (we assume $b > 0$) and $a(T)$ is temperature dependent. We can show that this system yields an appropriate phase transition if we allow $a(T)$ to change sign at the transition temperature T_C . Thus in the region of interest, near the transition, we write $a(T) = a_0(T - T_C)$ where a_0 is a positive constant. To find the ground state of the system, it is necessary to minimize the free energy so we look for solutions of $\partial F/\partial M = 0$. This condition implies

$$2M[a_0(T - T_C) + 2bM^2] = 0. \quad (6.2)$$

The left-hand side of this equation is a product of two terms, so either of them could be zero. This means

$$M = 0 \quad \text{or} \quad M = \pm \left[\frac{a_0(T_C - T)}{2b} \right]^{1/2}. \quad (6.3)$$

The second condition is only valid when $T < T_C$, otherwise one is trying to take the square root of a negative number. The first condition applies above or below T_C but below T_C it only produces a position of unstable equilibrium (which can be deduced by evaluating $\partial^2 F/\partial M^2$). Thus the magnetization follows the curve shown in Fig. 6.6(b); it is zero for temperatures $T \geq T_C$ and is non-zero and proportional to $(T_C - T)^{1/2}$ for $T < T_C$.

Landau's approach to studying phase transitions is called a **mean-field theory** which means that it assumes that all spins 'feel' an identical average exchange field produced by all their neighbours. This field is proportional to the magnetization. This approach is identical to the Weiss model outlined in Section 5.1.1. Mean-field theories are the simplest type of theory that can be constructed to describe many different types of phase transition and give similar results in each case. They go under different names in different cases: e.g. Bragg-Williams theory for order-disorder transitions in alloys. Mean-field theories fail to explain the critical region accurately because the assumption that all regions of the sample are the same then becomes particularly misplaced.

Mean-field theories ignore **correlations** and **fluctuations** which become very important near T_C . Very near to the critical temperature, large fluctuations are seen in the order parameter. The critical region is characterized by fluctuations,

Lev D. Landau (1908–1968)

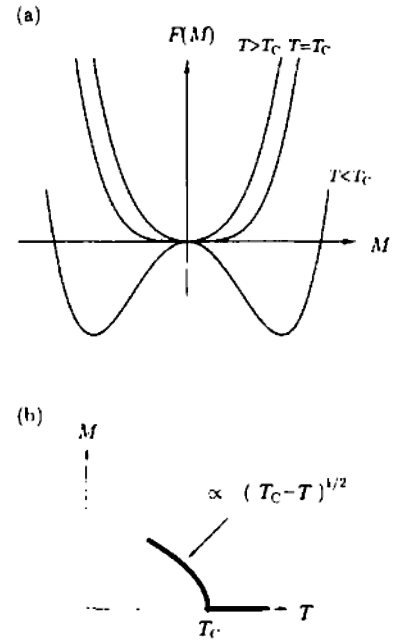


Fig. 6.6 (a) The free energy $F(M)$ of a ferromagnet. (b) The corresponding magnetization as a function of temperature.

¹ A visual demonstration of this phenomenon is **critical opalescence**, the blurring and clouding of images seen through a volume of gas near its critical point. This occurs because density fluctuations are strong near the critical point and give rise to large variations in refractive index.

at all lengthscales. When a saucepan of water is heated, the water warms quite quietly and unobtrusively until near the boiling point when it makes a great deal of noise and it bubbles and foams.¹ The same type of effect occurs near T_C in a ferromagnet. The dominant length scale which characterizes the fluctuations is the correlation length ξ . As $T \rightarrow T_C$, $\xi \rightarrow \infty$ i.e., the correlation length becomes infinite at the critical point. The importance of fluctuations is notable also from the form of $F(M)$. The curvature at $M = 0$, $(\partial^2 F / \partial M^2)_{M=0}$, tends to zero at $T = T_C$ so that the minimum at $M = 0$ becomes broad and flat-bottomed (see Fig. 6.6). Hence predictions about the critical region obtained from mean-field theories, though the theories are straightforward to solve, must be regarded with some caution.

6.2.2 Heisenberg and Ising models

An alternative approach to understanding the magnetic behaviour of solids is to consider particular microscopic models of the magnetic interaction. A commonly studied model is the **nearest neighbour Heisenberg model** which has a Hamiltonian

$$\hat{\mathcal{H}} = - \sum_{\langle ij \rangle} J \mathbf{S}_i \cdot \mathbf{S}_j, \quad (6.4)$$

where the constant J is the exchange integral and the symbol $\langle ij \rangle$ below the $\sum$ denotes a sum over nearest neighbours. The spins $\mathbf{S}_i$ are treated as three-dimensional vectors because we allow them to point in any direction in three-dimensional space. However the sum can be taken over a lattice of 1, 2 or 3 dimensions. Here it is important to distinguish between the dimensionality d of the lattice on which the spins sit and the dimensionality D of the spins themselves (in general D is known as the dimensionality of the *order parameter*). For the Heisenberg model $D = 3$ (because the spins are three-dimensional vectors). However we could be considering a lattice of these spins in 1, 2 or 3 dimensions (or 4 dimensions if we wanted to!) so d can be 1, 2, 3, ...

A related model is the **Ising model** in which the spins are only allowed to point up or down, i.e. we only consider the z component of the spin. The Hamiltonian of this model is

$$\hat{\mathcal{H}} = - \sum_{\langle ij \rangle} J S_i^z S_j^z. \quad (6.5)$$

Here the dimensionality of the order parameter D is equal to 1 (the spins are only allowed to point along $\pm \mathbf{z}$). Nevertheless we could arrange these one-dimensional spins on a lattice with $d = 1, 2, \dots$

6.2.3 The one-dimensional Ising model ($D = 1, d = 1$)

If the Ising spins are placed on a one-dimensional lattice, we will show that there is no phase transition. First, consider a chain with $N + 1$ spins (and hence we need to consider N 'bonds' between each neighbour). The Hamiltonian is

$$\hat{\mathcal{H}} = -2J \sum_{i=1}^N S_i^z S_{i+1}^z \quad (6.6)$$

Ernst Ising (1900–1998)

In Chapter 1, the operator for the z component of the spin was written as $\hat{S}_z$. Because we now also need to label the site of the spin, we will write the z as a superscript and put the site label as a subscript. Hence $\hat{S}_i^z$ is the operator for the z component of the spin at site i .

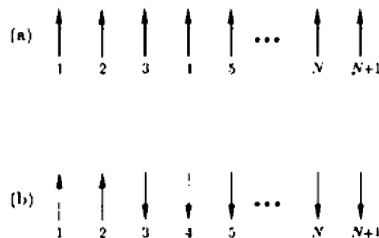


Fig. 6.7 The one-dimensional Ising model. (a) The ground state contains all $N + 1$ spins aligned ferromagnetically. (b) A single defect is added.

and assume $J > 0$ so that the ground state is obtained by having all adjacent spins lined up ferromagnetically (this is illustrated in Fig. 6.7(a)). The ground state thus has energy $-NJ/2$ because $S_i^z = +\frac{1}{2}$ for all values of z . Now consider adding one 'mistake', a single defect (see Fig. 6.7(b)). This costs an extra energy $E = J$ because we have to turn one favourable interaction (energy saving $J/2$) into an unfavourable one (energy cost $J/2$, so the change in energy is J). However, there is an entropy gain equal to $S = k_B \ln N$ because we can put the defect in any one of N places. As we let the chain get very large ($N \rightarrow \infty$) the energy cost of a defect remains the same (J) but the entropy gain becomes infinite. The properties are determined by the free energy $F = E - TS$ so that as long as the temperature is not zero, the entropy consideration means that the presence of the defect causes the free energy to plummet to $-\infty$. This means that defects can spontaneously form and in fact no long range order occurs for $T > 0$. Another way of saying this is that the critical temperature is zero. This consideration is valid for all models on one-dimensional lattices (because entropy always wins in one dimension) and we conclude that long range order is not possible in one-dimension.

6.2.4 The two-dimensional Ising model ($D = 1, d = 2$)

If the Ising spins are placed on a two-dimensional lattice, a phase transition to a magnetically ordered state below a non-zero critical temperature will result. This is because the energy cost and the entropy gain of making a defect both scale with the perimeter size of the defect (see Fig. 6.8). Energy and entropy therefore can have a fair fight with neither having an overwhelming advantage. The detailed solution of this model was one of the outstanding achievements of twentieth century statistical physics (Lars Onsager solved it in 1944) and his solution is beyond the scope of this book. This illustrates that even problems which are simple to state are by no means easy to solve.

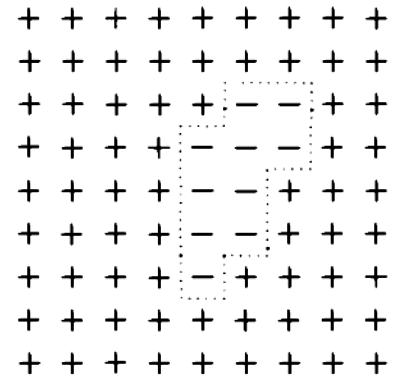


Fig. 6.8 The two-dimensional Ising model. Up spins are labelled + and down spins are labelled -. The energy and entropy of the down spin defect scale with the perimeter of the defect boundary.

Lars Onsager (1903–1976)

6.3 Consequences of broken symmetry

When you break symmetry there are various consequences:

- **Phase transitions:** The system will have a sharp change of behaviour at a temperature T_c . We describe this by saying the system has changed *phase* (e.g. liquid $\rightarrow$ solid, paramagnet $\rightarrow$ ferromagnet, etc). The region near the phase transition is called the *critical region* (see Section 6.4).
- **Rigidity:** Having broken the symmetry, the system will have a strong energetic preference for staying in that broken-symmetry state, and attempts by us to change the way the system has broken the symmetry meet with resistance. Thus crystals don't bend easily and ferromagnets show permanent magnetism (see Section 6.5).
- **Excitations:** At $T = 0$ the system is perfectly ordered. At finite temperature this order is weakened by excitations in the order parameter. In crystals these excitations are called lattice waves, quantized into phonons. In ferromagnets the analogous modes are called spin waves, quantized into magnons (see Section 6.6).

- **Defects:** If symmetry is broken differently in two adjacent parts of a macroscopic sample, the boundary will contain a defect: e.g. a dislocation in a crystal or a domain wall in a ferromagnet (see Section 6.7).

A summary of the properties of a number of different broken symmetry phases is contained in Table 6.1. Each is characterized by a high temperature disordered state and a low temperature ordered state. Each has an order parameter, a rigidity phenomenon, a set of excitations representing wave-like departures from the ordered state, and defects associated with breaking the symmetry differently in different spatial regions.

A crystalline solid gives rise to Bragg peaks in a diffraction experiment. Thus we can use as an order parameter ρ_G , the Fourier component of the charge density corresponding to a spatial frequency equal to a reciprocal lattice vector G . The case of an antiferromagnet is similar to that of a ferromagnet, except that the order parameter is the magnetization on one sublattice of the antiferromagnet. A superconductor has a complex order parameter given by $\psi = |\psi|e^{i\phi}$. Here $|\psi|^2$ represents the density of superconducting electrons (for a superfluid or Bose condensate, it represents the condensate fraction). The phase ϕ of the wave function is the broken symmetry; it is free to take any value at any point in the normal state, but in the superconducting state it becomes uniform over the whole sample. For a nematic liquid crystal (see Fig. 6.9), the order parameter is given by $S = \langle \frac{1}{2}(3 \cos^2 \theta - 1) \rangle$, which is the average value of the function $\frac{1}{2}(3 \cos^2 \theta - 1)$ where θ is the angle between the long axis of the molecule and the director, $\hat{n}$, a unit vector which points along the mean orientation of the axis of the liquid crystal molecule. An isotropic (liquid) state corresponds to $S = 0$. Full alignment corresponds to $S = 1$.

Table 6.1 The properties of broken symmetry phases. Here ρ_G is the Fourier component of the charge density corresponding to a spatial frequency equal to a reciprocal lattice vector G . The complex wave function in a superconductor is $\psi = |\psi|e^{i\phi}$. The electric polarization P is the electric dipole moment per unit volume.

Phenomenon	High T Phase	Low T Phase	Order parameter	Excitations	Rigidity phenomenon	Defects
crystal	liquid	solid	ρ_G	phonons	rigidity	dislocations, grain boundaries
ferromagnet	paramagnet	ferromagnet	M	magnons	permanent magnetism	domain walls
antiferromagnet	paramagnet	antiferromagnet	M (on sublattice)	magnons	(rather subtle)	domain walls
nematic (liquid crystal)	liquid	oriented liquid	$S = \langle \frac{1}{2}(3 \cos^2 \theta - 1) \rangle$	director fluctuations	various	disclinations, point defects
ferroelectric	non-polar crystal	polar crystal	P	soft modes	ferroelectric hysteresis	domain walls
superconductor	normal metal	superconductor	$ \psi e^{i\phi}$	—	superconductivity	flux lines

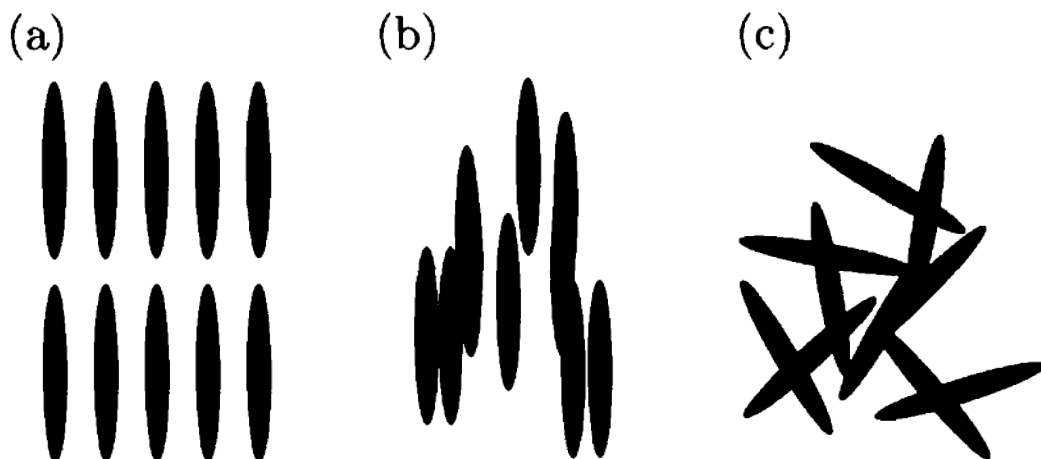


Fig. 6.9 A nematic liquid crystal contains rod-shaped molecules. (a) At low temperatures these order as a solid. (c) At high temperatures they are in a liquid state. (b) There is an intermediate temperature region where they are in the nematic state, in which the long-axis of the molecules become aligned, even though the positions of the centre of mass of the molecules are not ordered.

Table 6.2 Critical exponents for various models.

Model	Mean-field	Ising	Ising	Helsenberg
D	any	1	1	3
d	any	2	3	3
β	$\frac{1}{2}$	$\frac{1}{8}$	0.326	0.367
γ	1	$\frac{7}{4}$	1.2378(6)	1.388(3)
δ	3	15	4.78	4.78

6.4 Phase transitions

In Section 6.2 the Landau theory of ferromagnetism was presented. This is a mean-field theory in which an identical exchange field is felt by all spins. This leads to the magnetization behaving as $(T_C - T)^{1/2}$ below the transition. In real systems it is found that the magnetization does behave as $(T_C - T)^\beta$ close to the transition, but the exponent β is not necessarily equal to $\frac{1}{2}$. The exponent therefore gives important information about the nature of the phase transition. A number of other similar exponents, known as **critical exponents**, can be defined. Thus near the phase transition temperature T_C it is found experimentally that

$$\begin{array}{lll}
 \chi & \propto & (T - T_C)^{-\gamma} & T > T_C \\
 M & \propto & (T_C - T)^\beta & T < T_C \\
 M & \propto & H^{1/\delta} & T = T_C
 \end{array} \quad (6.7)$$

where β , γ , and δ are the critical exponents. The values taken by these critical exponents for various models are shown in Table 6.2.

As stated earlier, correlations and fluctuations are ignored in mean-field theories. This means that they cannot hope to give a correct description close to T_C , precisely where one might wish to calculate critical exponents. It turns out however that mean-field theory is a correct description of systems with four or more dimensions (i.e. $d \geq 4$) and predicts the correct critical exponents.

Despite the failure of mean-field theory to account successfully for critical behaviour in systems with dimensionality below four, it is only necessary

to consider a small representative set of ideal statistical models to calculate critical exponents of any physical system if the hypothesis of **universality** is accepted. This hypothesis is based on the observation that critical exponents do seem to be surprisingly independent of the type of phase transition, whether liquid–gas, ferromagnetic–paramagnetic, superconducting–non-superconducting, or any other. It is supposed that for a continuous phase transition, the critical exponents depend only on

- (1) The dimensionality of the *system*, d .
- (2) The dimensionality of the *order parameter*, D . (Actually the symmetry of the order parameter.)
- (3) Whether the forces are *short* or *long* range.

Hence, it is only important to look at particular **universality classes**, i.e. particular values of D and d for both short and long range forces, and pick the simplest models in each class and calculate the exponents for those. (Actually the assumptions stated above apply to static critical exponents, rather than the dynamic critical exponents which characterize time-dependent properties.)

We now need to list those models for which solutions are known. There are a number of exactly solved models which include the following:

- (1) All cases for $d = 1$. As we have seen, such systems do not exhibit continuous phase transitions.
- (2) All cases for $d \geq 4$, which give mean-field solutions.
- (3) All cases for long range interactions, which give mean-field solutions.
- (4) The case $d = 2$, $D = 1$. This is the 2D Ising model (see Table 6.2).
- (5) The case $D = \infty$ for any d . This is known as the spherical model.

Unfortunately, most real situations correspond to $d = 3$ and short-range interactions, which have not been solved exactly.

A method of calculating critical behaviour, even in models which cannot be exactly solved, was discovered in the early 1970s, in which a block of spins with volume L^d is considered, where L is much less than the correlation length ξ . If we increase (scale) the linear dimensions of this block by a factor n , the Hamiltonian will be suitably scaled, and it is possible to examine how the order parameter is thus changed (renormalized) by this transformation. If such a transformation is denoted by τ , then the Hamiltonian $\mathcal{H}$ transforms as $\mathcal{H}' = \tau(\mathcal{H})$. At the critical point, the correlation length $\xi \rightarrow \infty$, so that there must exist a limiting function $\mathcal{H}^*$ such that $\mathcal{H}^*$ is a fixed point of τ , i.e. $\tau(\mathcal{H}^*) = \mathcal{H}^*$. This approach, namely that of looking for fixed points of scaling transformations, is known as the **renormalization group method**, since the set of scaling transformations τ form a group. It has proved to be particularly successful in condensed matter physics, and has deep connections with renormalization in quantum field theory. A more detailed treatment of the renormalization group is however beyond the scope of this book.

Cluster models can be useful in describing the paramagnetic regime of ferromagnets, i.e. the region $T > T_C$. They retain some of the simplicity (and hence predictive power) of mean-field approaches but make some attempt to model the fluctuations. The idea is to evaluate the configuration of a single finite-size cluster of spins exactly, and then to couple that to the mean field from the rest of the sample. Of course the correlation length diverges at T_C and

so the size of the cluster needs to increase as you get closer to the transition. Very good agreement with data can be obtained if a cluster model is chosen in which the size of the clusters is unrestricted, and is allowed to vary.²

6.5 Rigidity

Breaking symmetry involves a choice of ground state: should the spins all point up, or down, or left or right? The energy of a macroscopic sample is minimized when the symmetry is broken the same way throughout its volume. If you try to make different parts of a macroscopic sample break symmetry differently, forces will appear reflecting an additional energy cost. This gives rise to a generalized rigidity or stiffness. This rigidity is intimately connected to the order, as can be deduced from the following argument for a crystal.

Consider a cubic crystal with lattice parameter a and imagine stretching one plane of bonds by increasing a to $a + u$ inducing a strain $\epsilon = u/a$ (see Fig. 6.10). This produces a stress equal to $G\epsilon$ (where G is an elastic modulus) so that the force on an individual bond is equal to $G\epsilon a^2 = Gau$. The energy stored in this bond is then $\frac{1}{2}Gau^2$. At high temperatures (large compared with the Debye temperature³) the mean stored energy in each bond is equal to $\frac{1}{2}k_B T$ (by the equipartition theorem, see Appendix E) so that

$$\langle u^2 \rangle = \frac{k_B T}{Ga} \quad (6.8)$$

and hence $\langle u^2 \rangle \propto 1/G$. Thus if G vanishes, the fluctuations diverge and *vice versa*. Hence the existence of a non-zero elastic modulus (i.e. a rigidity) is connected with the finiteness of the fluctuations (i.e. the stability of the crystal).

This means that because we have an ordered crystal, we must have a rigid crystal. Crystals do not deform easily (in contrast to liquids) because there is an elastic energy proportional to the elastic modulus and to $(\nabla u)^2$ where u is the lattice displacement, and the lattice transmits this force to the other end of a macroscopic sample. Solids are therefore rigid.

These ideas are also applicable to ferromagnets. The spins are aligned in a ferromagnet and it costs energy to turn them with respect to each other. Hence we have the phenomenon of permanent magnetism. If the magnetization is non-uniform there is an exchange cost proportional to $(\nabla \mathbf{M})^2$ (see Section 4.2.7, and eqn 4.18 in particular).

As a final example, consider a superconductor in which the phase of the wave function, ϕ , is the order parameter. In a superconductor, ϕ is uniform across a sample and there is a 'phase rigidity' energy proportional to $(\nabla \phi)^2$. (In fact twisting this phase across a sample produces a supercurrent $\propto \nabla \phi$.)

6.6 Excitations

A solid is ordered at $T = 0$, although zero-point fluctuations mean that, even then, atoms are not purely static. At non-zero temperature, the order is disrupted by thermally excited lattice vibrations, which are quantized as phonons. The behaviour of the phonons is characterized by a dispersion relation, i.e.

²See R. V. Chamberlin, *Nature*, **408**, 337 (2000).

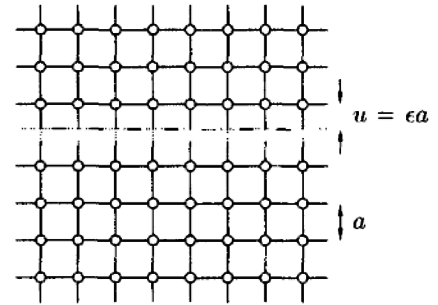


Fig. 6.10 A cubic crystal of lattice parameter a is stretched producing a strain $\epsilon = u/a$.

³The Debye temperature Θ_D gives the energy of the highest occupied phonon mode, divided by k_B , under the assumption that the phonons are non-dispersive. Of course there is always some dispersion, but Θ_D nevertheless characterizes a typical phonon energy of the system. Most Debye temperatures are in the range 100–1000 K. When $T \gg \Theta_D$, the phonon modes can be treated classically.

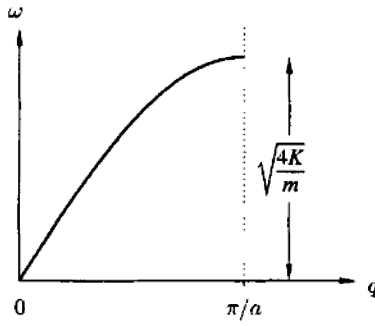


Fig. 6.11 The phonon dispersion for a one-dimensional monatomic chain of atoms of mass m connected by bonds with spring constant K . These phonons are acoustic. There is no gap ($\omega = 0$ at $q = 0$) and at small q , $\omega \approx v_s q$ where v_s is the speed of sound. The dispersion is massless because $\omega = 0$ at $q = 0$ (i.e. energy $E = \hbar\omega = 0$ at momentum $p = \hbar q = 0$). A relativistic particle has energy $E^2 = p^2 c^2 + m^2 c^4$, so that the energy at zero momentum is equal to mc^2 , i.e. proportional to the mass.

a relationship between angular frequency ω and wave vector q (equivalently between energy $\hbar\omega$ and momentum $\hbar q$). An example for a one-dimensional monatomic chain is shown in Fig. 6.11. The crucial feature of this is that $\omega = 0$ at $q = 0$ for acoustic phonons, so that it costs a vanishingly small energy to produce a phonon of wave vector q , provided its wavelength $\lambda = 2\pi/q$ is long enough. The fact that an acoustic phonon can be thermally generated, as long as the temperature is non-zero, is because there is no energy gap to leap across from the ground state of the system ($q = 0, \omega = 0$) to the lowest acoustic phonon level. There is no energy gap in the phonon dispersion relation at $q = 0$ (in contrast to the case for optic phonons). Acoustic phonons give rise to a T^3 heat capacity at low temperature in a three-dimensional crystal.

Whenever you have broken a continuous global symmetry (as you do when you make a solid from a liquid or a ferromagnet from a paramagnet) it is possible to produce long-wavelength excitations in the order parameter for vanishingly small energy cost. Such excitations are called **Goldstone modes** (or sometimes Goldstone bosons). Because they cost no energy they are 'massless'. In particle physics an example of this is the photon which is a Goldstone boson. For a superconductor, the situation is rather different because it turns out that you are breaking a continuous *local* symmetry and you don't get Goldstone modes. The reason for this is rather subtle and is connected with the Higgs mechanism.

A ferromagnet is perfectly ordered at $T = 0$ but at non-zero temperature the order is disrupted by **spin waves**, quantized as **magnons**. The crucial feature of these is that it costs a vanishingly small energy to produce a magnon, provided its wavelength is long enough. Thus the magnons play the same rôle in ferromagnets as phonons do in solids and are the Goldstone modes of the system. We will show below that, for an isotropic ferromagnet, there is no energy gap in the magnon dispersion relation at $q = 0$.

6.6.1 Magnons

In this section we will derive the magnon dispersion relation for an isotropic ferromagnet. This is an important problem so two alternative derivations will be presented, the first using a semiclassical approach and the second using a quantum mechanical approach.

- (I) We begin with a semiclassical derivation of the spin wave dispersion. First, recall the Hamiltonian for the Heisenberg model,

$$\hat{\mathcal{H}} = - \sum_{\langle ij \rangle} J \hat{\mathbf{S}}_i \cdot \hat{\mathbf{S}}_j \quad (6.9)$$

(which is eqn 6.4). In a one-dimensional chain each spin has two neighbours, so the Hamiltonian reduces to

$$\hat{\mathcal{H}} = -2J \sum_i \hat{\mathbf{S}}_i \cdot \hat{\mathbf{S}}_{i+1}. \quad (6.10)$$

We can calculate the time dependence of $\langle \hat{\mathbf{S}}_j \rangle$ using eqn C.7, with the

result that

$$\frac{d\langle \hat{S}_j \rangle}{dt} = \frac{1}{i\hbar} \langle [\hat{S}_j, \hat{H}] \rangle \quad (6.11)$$

$$= -\frac{2J}{i\hbar} \langle [\hat{S}_j, \dots + \hat{S}_{j-1} \cdot \hat{S}_j + \hat{S}_j \cdot \hat{S}_{j+1} + \dots] \rangle \quad (6.12)$$

$$= -\frac{2J}{i\hbar} \langle [\hat{S}_j, \hat{S}_{j-1} \cdot \hat{S}_j] + [\hat{S}_j, \hat{S}_j \cdot \hat{S}_{j+1}] \rangle \quad (6.13)$$

$$= \frac{2J}{\hbar} \langle \hat{S}_j \times (\hat{S}_{j-1} + \hat{S}_{j+1}) \rangle. \quad (6.14)$$

We now treat the spins at each site as classical vectors. The ground state of the system has all the spins aligned, say along the z axis, so that $S_j^z = S$, $S_j^x = S_j^y = 0$. Consider a state which is a small departure from this state with $S_j^z \approx S$, $S_j^x, S_j^y \ll S$, so that

$$\frac{dS_j^x}{dt} \approx \frac{2JS}{\hbar} (2S_j^y - S_{j-1}^y - S_{j+1}^y) \quad (6.15)$$

$$\frac{dS_j^y}{dt} \approx -\frac{2JS}{\hbar} (2S_j^x - S_{j-1}^x - S_{j+1}^x) \quad (6.16)$$

$$\frac{dS_j^z}{dt} \approx 0. \quad (6.17)$$

We now look for normal mode solutions, so put

$$S_j^x = A e^{i(qja - \omega t)} \quad (6.18)$$

$$S_j^y = B e^{i(qja - \omega t)} \quad (6.19)$$

where q is a wave vector. A little algebra leads to $A = iB$ (showing that the x and y motion are $\pi/2$ out of phase) and hence that

$$\hbar\omega = 4JS(1 - \cos qa), \quad (6.20)$$

which is the dispersion relation for the spin waves and which is plotted in Fig. 6.12. Thus the spin waves are as depicted in Fig. 6.13.

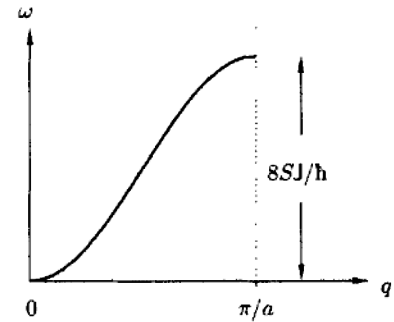


Fig. 6.12 The spin wave dispersion relation for a one-dimensional chain of spins.

(a)



(b)



Fig. 6.13 A spin wave on a line of spins. (a) perspective view. (b) view from above.

- (2) It is perhaps more satisfying to derive eqn 6.20 using a quantum mechanical method, and so we now repeat the derivation.⁴ We consider the ground state of the system, $|\Phi\rangle$, which consists of all the spins lying along the $+z$ direction. In one dimension the Hamiltonian for the Heisenberg model can be written

$$\hat{H} = -2J \sum_i \left[\hat{S}_i^z \hat{S}_{i+1}^z + \frac{1}{2} (\hat{S}_i^+ \hat{S}_{i+1}^- + \hat{S}_i^- \hat{S}_{i+1}^+) \right], \quad (6.21)$$

⁴We take $S = \frac{1}{2}$ for now, but it can be easily generalized.

$$|\Phi\rangle = \cdots \uparrow\uparrow\uparrow\uparrow\uparrow\uparrow\uparrow\uparrow\uparrow \cdots$$

j

$$|j\rangle = \cdots \uparrow\uparrow\downarrow\uparrow\uparrow\uparrow\uparrow\uparrow\uparrow \cdots$$

Fig. 6.14 The state $|\Phi\rangle$ consists of all the spins lying along the $+z$ direction. The state $|j\rangle$ is the ground state with the spin at site j flipped.

so that $\hat{\mathcal{H}}|\Phi\rangle = -NS^2|\Phi\rangle$. Now to create an excitation, flip a spin at site j , so let us consider a state $|j\rangle = \hat{S}_j^-|\Phi\rangle$ which is the ground state with the spin at site j flipped (see Fig. 6.14). By flipping a spin, we have changed the total spin of the system by $\frac{1}{2} - (-\frac{1}{2}) = 1$. This excitation therefore has integer spin and is a boson. If we apply the Hamiltonian to this new state, we get

$$\hat{\mathcal{H}}|j\rangle = 2 \left[(-NS^2J + 2SJ)|j\rangle - SJ|j+1\rangle - SJ|j-1\rangle \right], \quad (6.22)$$

which is not a constant multiplied by $|j\rangle$, so this state is not an eigenstate of the Hamiltonian. Nevertheless, we can diagonalize the Hamiltonian by looking for plane wave solutions of the form

$$|q\rangle = \frac{1}{\sqrt{N}} \sum_j e^{iqR_j} |j\rangle. \quad (6.23)$$

The state $|q\rangle$ is essentially a flipped spin delocalized (smeared out) across all the sites. Since it is composed of states representing a single flipped spin, the total spin of $|q\rangle$ itself has the value $NS - 1$. It is then straightforward to show that

$$\hat{\mathcal{H}}|q\rangle = E(q)|q\rangle, \quad (6.24)$$

where

$$E(q) = -2NS^2J + 4JS(1 - \cos qa). \quad (6.25)$$

The energy of the excitation is then $\hbar\omega = 4JS(1 - \cos qa)$ which is the same as the result we had above in eqn 6.20.

6.6.2 The Bloch $T^{3/2}$ law

At small q , eqn 6.20 yields

$$\hbar\omega \approx 2JSq^2a^2, \quad (6.26)$$

so that $\omega \propto q^2$. In three dimensions, the density of states is given by

$$g(q) dq \propto q^2 dq \quad (6.27)$$

which leads to

$$g(\omega) d\omega \propto \omega^{1/2} d\omega \quad (6.28)$$

at low temperature where only small q and small ω are important. The spin waves are quantized in the same way as lattice waves. The latter are termed phonons, and so in the same way the former are termed magnons. As shown in Section 6.6.1, magnons have a spin of one and are bosons.

The number of magnon modes excited at temperature T , n_{magnon} , is calculated by integrating the magnon density of states over all frequencies after multiplying by the Bose factor, $(\exp(\hbar\omega/k_B T) - 1)^{-1}$, which must be included because magnons are bosons. Thus the result is given by

$$n_{\text{magnon}} = \int_0^\infty \frac{g(\omega) d\omega}{\exp(\hbar\omega/k_B T) - 1}, \quad (6.29)$$

which can be evaluated using the substitution $x = \hbar\omega/k_B T$. At low temperature, where $g(\omega) \propto \omega^{1/2}$ in three dimensions, this yields the result

$$n_{\text{magnon}} = \left(\frac{k_B T}{\hbar} \right)^{3/2} \int_0^\infty \frac{x^{1/2} dx}{e^x - 1} \propto T^{3/2}. \quad (6.30)$$

Since each magnon mode which is thermally excited reduces the total magnetization⁵ by $S = 1$, then at low temperature the reduction in the spontaneous magnetization from the $T = 0$ value is given by

$$\frac{M(0) - M(T)}{M(0)} \propto T^{3/2}. \quad (6.31)$$

This result is known as the Bloch $T^{3/2}$ law and it fits experimental data in the low temperature regime (see Fig. 6.15).⁶ The energy of the magnon modes is given by

$$E_{\text{magnon}} = \int_0^\infty \frac{\hbar\omega g(\omega) d\omega}{\exp(\hbar\omega/k_B T) - 1} \propto T^{5/2}, \quad (6.32)$$

so that the heat capacity $C = \partial E_{\text{magnon}}/\partial T$ is also proportional to $T^{3/2}$.

⁵ Because each magnon mode is a delocalized single reversed spin, see the previous section.

⁶ The Bloch $T^{3/2}$ law is only really correct for the spontaneous magnetization within a domain. The data shown in Fig. 6.15 actually measure this because they were obtained using μSR which measures the local internal field with zero applied field.

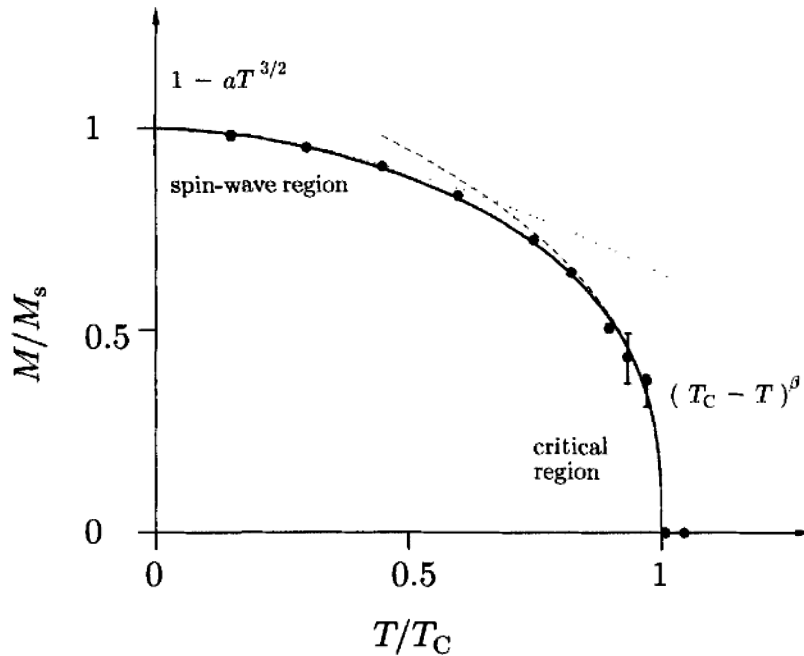


Fig. 6.15 The spontaneous magnetization in a ferromagnet. At low temperatures this can be fitted using the spin-wave model and follows the Bloch $T^{3/2}$ law. Near the critical temperature, the magnetization is proportional to $(T - T_C)^\beta$ where β is a critical exponent. Neither behaviour fits the real data across the whole temperature range. The data are for an organic ferromagnet which has $T_C \approx 0.67$ K for which $\beta \approx 0.36$, appropriate for the three-dimensional Heisenberg model.

6.6.3 The Mermin–Wagner–Berezinskii theorem

In one and two dimensions, however, the integral in eqn 6.29 diverges, so that $M \rightarrow 0$ for all $T > 0$ for the isotropic 1-D and 2-D Heisenberg models.⁷ The number of spin waves which are generated at finite temperature diverges, and hence spontaneous ferromagnetism is not possible. This result was first proved by Mermin and Wagner in 1966 and independently by Berezinskii. The absence of long range order in two-dimensional systems (with a continuous symmetry) is often referred to as the Mermin–Wagner–Berezinskii theorem.

⁷ For different dimensions, the function $g(\omega)$ in eqn 6.29 has a different dependence on ω .

⁸If the wavelength of the spin wave is sufficiently long, the exchange energy cost $\propto \int (|\nabla M_x|^2 + |\nabla M_y|^2) dx dy$ (in two dimensions) is minimized. See Exercise 6.7.

⁹This idea is explored in Exercise 6.7

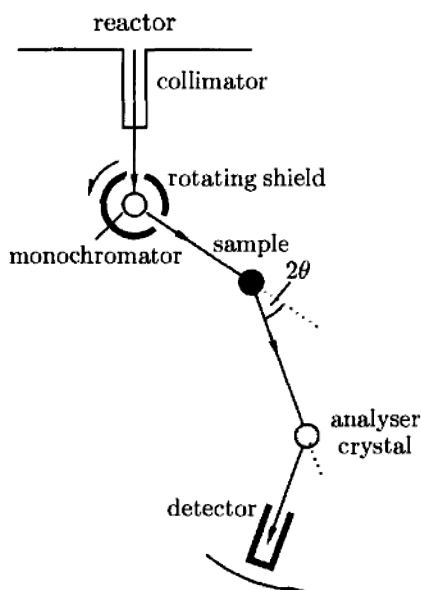


Fig. 6.16 Schematic diagram of a neutron triple-axis spectrometer. By setting the angle of the rotating shield and the monochromator crystal, the incident neutron energy can be selected. The sample can also be rotated. The analysing crystal can also be rotated, allowing the scattered neutron energy to be measured.

¹⁰The addition of $\mathbf{G}$ is necessary because the dispersion relation of magnons, like that of phonons, is periodic in the reciprocal lattice.

This result only applies to an isotropic Heisenberg ferromagnet. This possesses rotational symmetry so that all of the spin directions can be globally rotated without any additional energy cost. This means that long wavelength excitations, in which the spin state may deviate from its ground state value over a considerable distance, cost very little energy.⁸ Thus a fluctuation of the spins can be excited with very little energy cost; in one and two dimensions they destroy the long range order. If, however, there is significant anisotropy, there will be an energy cost associated with rotating the spins from their ground state value.

It turns out that the anisotropy energy penalty incurred by allowing these fluctuations increases with the square of R , the radius of the excitation, and hence the anisotropy energy will suppress all but the smallest of these non-linear fluctuations.⁹ It is the presence of such symmetry-breaking fields which can stabilize long range order in two-dimensional systems. There is also a dipolar interaction between spins in real systems which, although much weaker than the exchange interaction, is anisotropic and can act in a similar way to suppress the growth of fluctuations.

It is observed in experiment that some ultra-thin magnetic films can show spontaneous ferromagnetism. Sometimes the anisotropy is such that it is energetically favourable for the spins to be perpendicular to the plane of the film. (The system is said to have a perpendicular easy axis for magnetization.) In this case the anisotropy leads to a gap in the spin wave excitation spectrum (between the ground state and the excited states) and long range magnetic order can exist at finite temperature. Dipolar interactions, or anisotropy of the energy of the spins in the plane of the film, can lead to a similar effect and stabilize the magnetization.

6.6.4 Measurement of spin waves

Elastic neutron scattering, which is used for magnetic structure determination, was discussed in Section 5.7.2. Spin wave dispersions can be measured using a technique known as **inelastic neutron scattering**. The magnitude of the incident neutron wave vector $\mathbf{k}$ is now no longer equal to the magnitude of the scattered neutron wave vector $\mathbf{k}'$. The energy of the neutron also changes from

$$E = \frac{\hbar^2 k^2}{2m_n} \quad (6.33)$$

to

$$E' = \frac{\hbar^2 k'^2}{2m_n} \quad (6.34)$$

This is because the neutron has produced in the sample an excitation of energy $\hbar\omega$ and wave vector $\mathbf{q}$. Conservation of energy and momentum implies that

$$E = E' + \hbar\omega \quad (6.35)$$

$$\mathbf{k} = \mathbf{k}' + \mathbf{q} + \mathbf{G}, \quad (6.36)$$

where $\mathbf{G}$ is a reciprocal lattice wave vector,¹⁰ so that a measurement of $\mathbf{k}$, $\mathbf{k}'$, E and E' allows a determination of ω and $\mathbf{q}$. Neutrons have energies similar to the energies of atomic and electronic processes, i.e. in the meV to eV range.

Thus we can probe energy scales from the μeV of quantum tunnelling, through molecular translations, rotations, vibrations and lattice modes, to eV transitions within the electronic structure of materials; magnon energies are typically in the range 10^{-3} – 10^{-2} eV and therefore can be effectively measured using inelastic neutron scattering.

Experiments are typically performed using a **neutron triple-axis spectrometer** (see Fig. 6.16), so called because the angles of the monochromator, sample and analyser crystals can all be separately varied. This allows scattered neutrons corresponding to a large range of possible values of ω and $\mathbf{q}$ to be measured. In particular, it allows scans to be performed in which q varies with fixed ω (or vice versa). Example data obtained by this technique are shown in Figures 6.17 for $\text{Co}_{0.92}\text{Fe}_{0.08}$ and 6.18 for the ferromagnetic oxide $\text{La}_{0.7}\text{Pb}_{0.3}\text{MnO}_3$ (which exhibits colossal magnetoresistance, see Section 8.9.5).

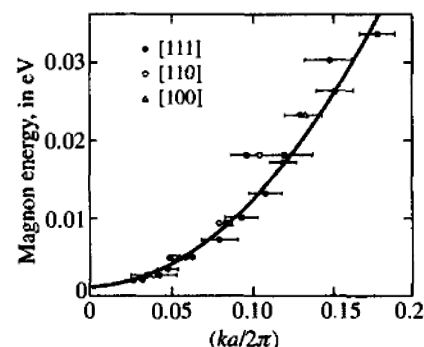
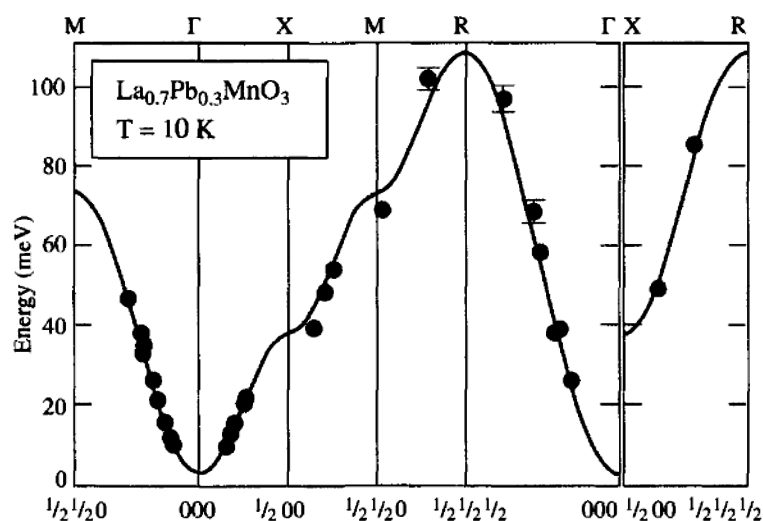


Fig. 6.17 The spectrum of magnon energy for spin waves in an alloy of $\text{Co}_{0.92}\text{Fe}_{0.08}$ obtained at room temperature (Sinclair and Brockhouse 1960). In this figure, k is the magnon wave vector corresponding to q in the text.

Fig. 6.18 Spin wave dispersion in the ferromagnetic oxide $\text{La}_{0.7}\text{Pb}_{0.3}\text{MnO}_3$ measured at 10 K by inelastic neutron scattering. The horizontal axis corresponds to the magnon wave vector. After Perring *et al.* 1996.

6.7 Domains

If different regions of a macroscopic system break symmetry in different ways, then in the interface between these regions the rigidity can break down. In general we expect domain walls, defects, vortices, dislocations and other singularities. In ferromagnets the most important singularity is the domain wall.

Weiss first proposed that a ferromagnet contains a number of small regions called **domains**, within each of which the local magnetization reaches the saturation value. The direction of the magnetization of different domains need not be parallel. Domains are separated by **domain walls**. The existence of domains explains the surprising observation that in some ferromagnetic specimens¹¹ it can be possible, at room temperature, to attain saturation magnetization of the whole sample (corresponding to $\mu_0 M \sim 1$ T) by the application of a very weak magnetic field (as low as 10^{-6} T). Such low applied fields would

¹¹ Which are magnetically 'soft', see Section 6.7.9.

¹²If $\chi \sim 10^{-3}$ for a particular paramagnet, see Table 2.1, then an applied field of 10^{-6} T would produce a magnetization of $\mu_0 M \sim 10^{-9}$ T.

¹³As we shall see, in some ferromagnetic materials the movement of domain walls can in fact be energetically costly. Such substances would require large applied magnetic fields to switch them from one magnetic state to the other.

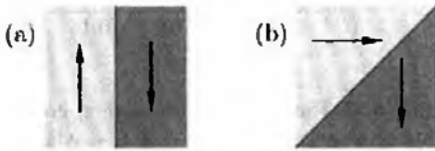


Fig. 6.19 (a) 180° domain wall. (b) 90° domain wall.

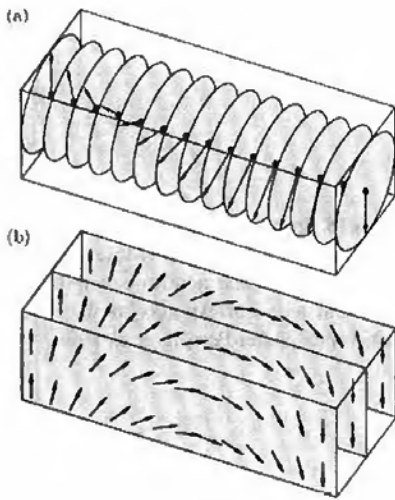


Fig. 6.20 (a) A Bloch wall. (b) A Néel wall.

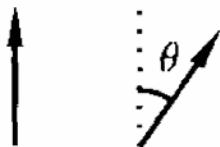


Fig. 6.21 Two spins at an angle θ with respect to each other, producing an energy cost $\frac{1}{2}JS^2\theta^2$.

¹⁴Using $\cos \theta \approx 1 - \theta^2/2$ for $\theta \ll 1$.

have negligible effect on a paramagnet.¹² The large effect in the ferromagnetic specimen is because the applied field does not have to order the magnetic moments macroscopically (they are all already ordered!) but merely to cause the domains to align. This can be achieved by the energetically painless process of domain wall motion.¹³

In the same ferromagnetic specimen, it is also possible for the magnetization to be zero in zero applied field. This is also a manifestation of domains; the magnetization is saturated within each domain, but the directions of the magnetization of each domain are such that the net magnetization of the specimen is zero.

6.7.1 Domain walls

Between adjacent domains there is a boundary called a domain wall. The domain walls can be classified according to the angle between the magnetization in the two domains (see Fig. 6.19). A 180° domain wall separates domains of opposite magnetization. A 90° domain wall separates domains of perpendicular magnetization.

The most common type of 180° wall is a Bloch wall (see Fig. 6.20(a)) in which the magnetization rotates in a plane parallel to the plane of the wall. Another possible configuration is the Néel wall (see Fig. 6.20(b)) in which the magnetization rotates in a plane perpendicular to the plane of the wall. Let us attempt to calculate the domain wall width in a Bloch wall.

In a ferromagnet it costs energy to rotate neighbouring spins. Two spins, S_1 and S_2 , which are at angle θ with respect to each other (Fig. 6.21) have an energy $-2JS_1 \cdot S_2 = -2JS^2 \cos \theta$. If $\theta = 0$, their energy is $-2JS^2$. Hence the energy cost of having $\theta \neq 0$ is approximately¹⁴ $JS^2\theta^2$ if $\theta \ll 1$. In a Bloch wall, spins rotate over N sites by an angle π (see Fig. 6.20(a)). Hence the energy cost of a line of spins is equal to N contributions of $JS^2\theta^2$ where $\theta = \pi/N$, i.e. to $JS^2\pi^2/N$. In a Bloch wall we have planes of spins (see Fig. 6.20) and so we are interested in σ_{BW} , the energy per unit area of the Bloch wall. In a square metre of wall, there are $1/a^2$ lines of spins like the one we have calculated. Hence

$$\sigma_{BW} = JS^2 \frac{\pi^2}{Na^2} \quad (6.37)$$

which tends to zero as $N \rightarrow \infty$. This result would seem to indicate that if a domain wall formed it would just unwind itself, growing in size throughout the entire system. This is because it costs energy to have spins twisted with respect to each other, and therefore they will all untwist unless some other interaction stops them. This other interaction is the magnetocrystalline anisotropy.

6.7.2 Magnetocrystalline anisotropy

Crystals possess a magnetic easy axis and a hard axis. Along certain crystallographic directions it is easy to magnetize the crystal, along others it is harder (this is shown for single crystals of Fe, Co and Ni in Fig. 6.22). In Co, for example, this anisotropy leads to an additional energy of the form

$$E = K_1 \sin^2 \theta + K_2 \sin^4 \theta \quad (6.38)$$

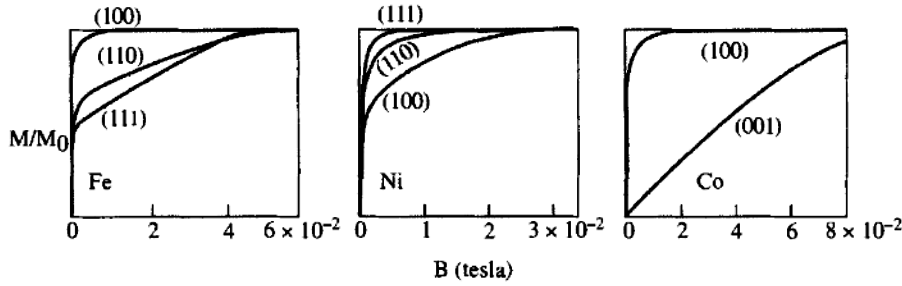


Fig. 6.22 Magnetization in Fe, Co and Ni for applied fields in different directions showing anisotropy. After Honda and Kaya 1926, Kaya 1928.

where K_1 and K_2 are anisotropy constants and θ is the angle between the magnetization and the stacking direction of the hexagonally close packed planes. Because these constants are positive the energy is minimized when the magnetization lies along the stacking direction. In eqn 6.38, E , K_1 and K_2 are energy densities (i.e. they are measured in J m^{-3}). The anisotropy constants are found to be strongly temperature dependent.

Equation 6.38 is appropriate for uniaxial anisotropy, in which the energy depends on the angle to a single axis (in Co this is the stacking axis of the hexagonally close packed planes). In a cubic system, the appropriate expression is

$$E = K_1(m_x^2m_y^2 + m_y^2m_z^2 + m_z^2m_x^2) + K_2m_x^2m_y^2m_z^2 + \dots, \quad (6.39)$$

where $\mathbf{m} = (m_x, m_y, m_z) = \mathbf{M}/|\mathbf{M}|$. In spherical coordinates this is

$$E = K_1 \left(\frac{1}{4} \sin^2 \theta \sin^2 2\phi + \cos^2 \theta \right) \sin^2 \theta + \frac{K_2}{16} \sin^2 2\phi \sin^2 2\theta \sin^2 \theta + \dots \quad (6.40)$$

The anisotropy energy arises from the spin-orbit interaction and the partial quenching of the angular momentum. Anisotropy energies are usually in the range 10^2 – 10^7 J m^{-3} . This corresponds to an energy per atom in the range 10^{-8} – 10^{-3} eV . The anisotropy energy is larger in lattices (of magnetic ions) of low symmetry and smaller in lattices of high symmetry. For example, cubic Fe and Ni have K_1 equal to $4.8 \times 10^4 \text{ J m}^{-3}$ and $-5.7 \times 10^3 \text{ J m}^{-3}$ respectively, but hexagonal Co has $K_1 = 5 \times 10^5 \text{ J m}^{-3}$. Low symmetry permanent magnet materials $\text{Nd}_2\text{Fe}_{14}\text{B}$ and SmCo_5 have K_1 equal to $5 \times 10^6 \text{ J m}^{-3}$ and $1.7 \times 10^7 \text{ J m}^{-3}$ respectively.

An additional energy term is due to the demagnetizing energy associated with the sample shape and is referred to as **shape anisotropy**. In thin films, a shape anisotropy $\frac{1}{2}\mu_0 M^2 \cos^2 \theta$ (where θ is the angle between the film normal and $\mathbf{M}$) leads to an energetic saving for keeping the magnetization in the plane of the film.

6.7.3 Domain wall width

In the magnetic domains of a ferromagnet the magnetization will prefer to lie along the easy direction but between domains, in the domain wall, it will have to rotate and a component will lie along the hard axis which will cost energy. If we assume a simple form for the anisotropy energy density, namely $E = K \sin^2 \theta$ where K is an anisotropy constant, then we can easily find an

expression for the anisotropy energy contribution to a Bloch wall. We take $K > 0$ so that spins prefer to line up along $\theta = 0$ or $\theta = \pi$. We add up the contribution from each of the N spins and replace the sum by an integral for simplicity (the continuum limit). Thus the energy density contribution is

$$\sum_{i=1}^N K \sin^2 \theta_i \approx \frac{N}{\pi} \int_0^\pi K \sin^2 \theta d\theta = \frac{NK}{2}. \quad (6.41)$$

The energy contribution per unit area of wall can then be written as $NKa/2$. Thus the total energy per unit area of the domain wall, including the contribution from the exchange energy (eqn 6.37) and the contribution from the anisotropy energy (eqn 6.41) is

$$\sigma_{BW} = JS^2 \frac{\pi^2}{Na^2} + \frac{NKa}{2}. \quad (6.42)$$

This gives us the required behaviour since we have two terms, one proportional to $1/N$ (tending to unwind the wall and make it bigger) and the other proportional to N (tending to tighten it and make it smaller). We find the equilibrium configuration using $dE_{BW}/dN = 0$ which leads to a value of N given by

$$N = \pi S \sqrt{2J/Ka^3} \quad (6.43)$$

so that the width of the Bloch wall is

$$\delta = Na = \pi S \sqrt{\frac{2J}{Ka}}, \quad (6.44)$$

where a is the lattice spacing. Larger J makes the wall thicker, larger K makes it smaller. The energy per unit area of the domain wall is

$$\sigma_{BW} = \pi S \sqrt{\frac{2JK}{a}}. \quad (6.45)$$

In terms of the parameter A defined in eqn 4.19 (for a cubic crystal) we have the domain wall thickness

$$\delta = \pi \sqrt{\frac{A}{K}}, \quad (6.46)$$

and the energy per unit area as

$$\sigma_{BW} = \pi \sqrt{AK}. \quad (6.47)$$

6.7.4 Domain formation

Since it costs this energy to make a domain wall, one might wonder why more than one domain ever forms in the first place. The reason is that the formation of domains saves energy associated with dipolar fields. Because $\nabla \cdot \mathbf{H} = -\nabla \cdot \mathbf{M}$ (see Appendix B) then whenever $\mathbf{M}$ stops and starts, at the edges of a sample for example, the magnetic field diverges and this produces demagnetizing fields which fill space and which cost $B^2/2\mu_0$ Joules of energy per cubic metre. The energy associated with the demagnetizing field is called, variously, the

demagnetization energy, magnetostatic energy or dipolar energy. It takes the value

$$-\frac{\mu_0}{2} \int_V \mathbf{M} \cdot \mathbf{H}_d d\tau, \quad (6.48)$$

where $\mathbf{H}_d$ is the demagnetizing field and the integral is taken over the volume of the sample (see Appendix D). For an ellipsoidally shaped sample magnetized along one of its principal axes, this energy reduces to

$$\frac{\mu_0}{2} N M^2 V \quad (6.49)$$

where N is the demagnetizing factor and V is the sample volume.

This dipolar energy can be saved by breaking the sample into domains, but each domain created costs energy because of the cost of the domain walls. Therefore, in the same way as the size of a domain wall is a balance between the exchange and anisotropy energies, so the formation of domains is a balance between the cost of a demagnetizing field and the cost of a domain wall.

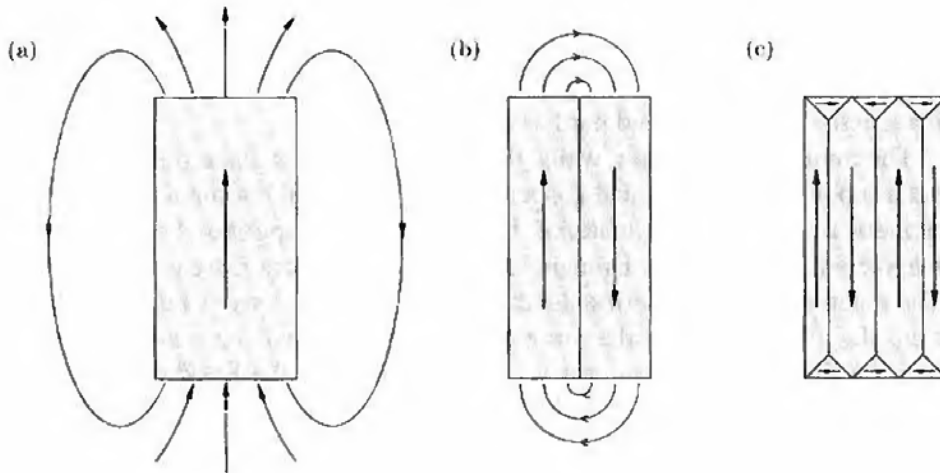


Fig. 6.23 A sample which is (a) uniformly magnetized, (b) divided into two domains, and (c) with a simple closure domain structure.

Figure 6.23 shows three different choices for the domain structure for a ferromagnetic sample. The single domain structure in Fig. 6.23(a) has no domain walls but a large dipolar energy. The dipolar energy can be reduced by breaking the sample into two domains as shown in Fig. 6.23(b), albeit at the cost of introducing a domain wall. The so-called **closure domain structure** in Fig. 6.23(c) eliminates the dipolar energy but introduces a number of domain walls.

Dipolar energy can also determine the type of domain wall that can form. The Bloch wall is favoured in the bulk because it leads to a smaller dipolar energy. The Néel wall tends to be favoured in thin films (where there is a dipolar energy cost to rotating the spins out of the plane of the film).

6.7.5 Magnetization processes

Figure 6.24 shows a typical hysteresis loop expected for measuring the magnetization as a function of the applied magnetic field for a ferromagnetic sample. If the sample is magnetized to the saturation magnetization M_s by an applied field, then when the applied field is reduced to zero the magnetization

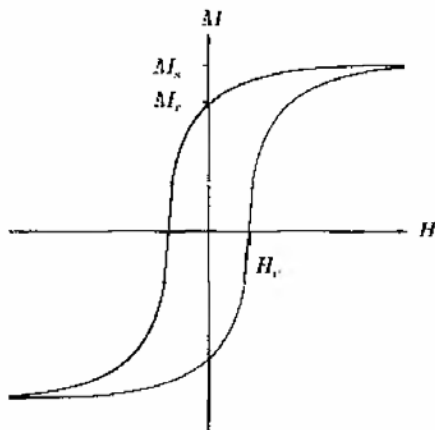


Fig. 6.24 A hysteresis loop showing the saturation magnetization M_s , the remanent magnetization M_r and the coercive field H_c .

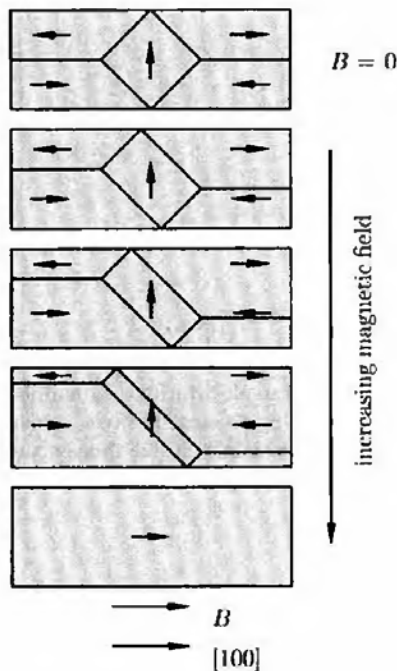


Fig. 6.25 Effect of an applied field on the domain pattern on the surface of a single crystal iron whisker showing domain wall displacement, as the applied B field increases from 0 up to a maximum value.

Heinrich Barkhausen (1881–1956)

John Kerr (1824–1907)

reduces to the remanent magnetization M_r . A magnetic field equal to the coercive field H_c is needed to switch the magnetization into the opposite direction. The parameters M_r and H_c can be used to characterize a ferromagnet.

The length of a ferromagnet changes slightly when it is magnetized. This effect is known as magnetostriction and demonstrates that the magnetic and elastic behaviour of a material are connected (the connection is called magnetoelastic coupling). The crystal deforms because it can lower its anisotropy energy which can depend on the strain of the crystal. For example, in a cubic system it may be energetically favourable for the crystal to deform slightly from exactly cubic symmetry if doing so will save more anisotropy energy than it will cost in elastic energy.

When a demagnetized ferromagnet is magnetized, various processes occur. First is domain-wall motion; domains which are aligned favourably with respect to the applied magnetic field grow at the expense of domains which are unfavourably aligned. (This process is shown in Fig. 6.25.) At higher fields, domain rotation can occur in which the anisotropy energy can be outweighed and the magnetization can suddenly rotate away from the original direction of magnetization to the crystallographic easy axis which is nearest to the field direction. The final domain process at highest magnetic fields is coherent rotation of the domains to a direction aligned with the magnetic field, irrespective of the easy and hard axes.

The motion of domain walls through a magnetic material depends in detail upon the metallurgical properties of the material. Domain walls can be pinned by strains in the material, by surfaces and impurities because of the magnetoelastic coupling. Domain wall pinning therefore increases coercivity. The magnetization of a ferromagnet also changes by a series of discontinuous steps due to domain boundary motion, so that very small steps are sometimes seen on the magnetization curves. This is known as the **Barkhausen effect**. Low level acoustic emission also sometimes accompanies the magnetization process because of these sudden discontinuous changes in magnetization and their coupling to the elastic modes of the system. This effect is known as magnetoacoustic emission.

6.7.6 Domain wall observation

Domains can be observed using a variety of techniques. In the **Bitter powder technique** a colloidal suspension of fine magnetic particles is placed on the surface of the sample. The magnetic particles collect near domain walls where there are strong local fields which attract them. These can be observed using optical microscopy. It is also possible to use polarized light reflecting from a polished magnetized sample to observe domains, a technique known as **Kerr microscopy**. The incident beam of light is polarized and its plane of polarization is rotated by reflection from the magnetized sample (this is the Kerr effect, see Section 8.8). The degree of rotation can be measured using a second polarizer (the analyser). If a beam of light is focussed to a spot and scanned across the surface of a magnetic sample, then for a particular setting of the analyser polarizer, some domains will appear bright and others dark. If the sample is transparent, the experiment can be performed in transmission (the Faraday effect).

Electron microscopy can also be effective as the electron beam is deflected by the Lorentz force due to the internal field. Boundaries between domains (across which this deflection can suddenly change in sign as the beam is scanned over) show up particularly well. This technique is called **Lorentz microscopy**.

In the technique of **magnetic force microscopy**¹⁵ a tiny cantilever with a magnetized tip is scanned across a sample surface. The stray fields produced by a domain wall cause the cantilever to bend, and this bending (detected by, for example, a change in the resonant frequency of the cantilever) is used to image the surface.

¹⁵ A magnetic force microscope is based on the well-known atomic force microscope.

6.7.7 Small magnetic particles

In many ferromagnetic samples the lowest energy state at zero applied field is the demagnetized state, so that the overall magnetic moment is zero. If the sample size is reduced, surface energies (such as the domain wall energies) become progressively more costly in comparison with volume energies (such as the demagnetizing energy).¹⁶ Thus a critical dimension may be reached below which it is energetically favourable to remove the domain walls so that the sample consists of a single magnetized domain. It would thus behave like a small permanent magnet.

¹⁶ This is because surface energies scale as the (sample size)² whereas volume energies scale as the (sample size)³.

In the following example, we estimate the critical size for a single domain particle. The calculation will assume that this critical size is still larger than the domain wall width.



Example 6.1

Consider a ferromagnetic particle of radius r . The energy of the single domain state (Fig. 6.26(a)) is, using eqn D.12 and $N = \frac{1}{3}$ for a sphere, given by

The volume of a sphere is $V = \frac{4}{3}\pi r^3$.

$$E_{(a)} = \frac{1}{6}\mu_0 M^2 V = \frac{2}{9}\mu_0 \pi M^2 r^3. \quad (6.50)$$

The state in Fig. 6.26(b), appropriate for cubic anisotropy, approximately removes all demagnetizing energy, but introduces 90° domain walls of area $2 \times \pi r^2$, leading to an energy cost

$$E_{(b)} = 2\pi r^2 \sigma_w^{90^\circ}, \quad (6.51)$$

where $\sigma_w^{90^\circ}$ is the energy cost of a 90° domain wall per unit area. The state in Fig. 6.26(a) thus becomes more favourable than the state in Fig. 6.26(b) when $E_{(a)} < E_{(b)}$, i.e. when

$$r < \frac{9\sigma_w^{90^\circ}}{\mu_0 M^2}. \quad (6.52)$$

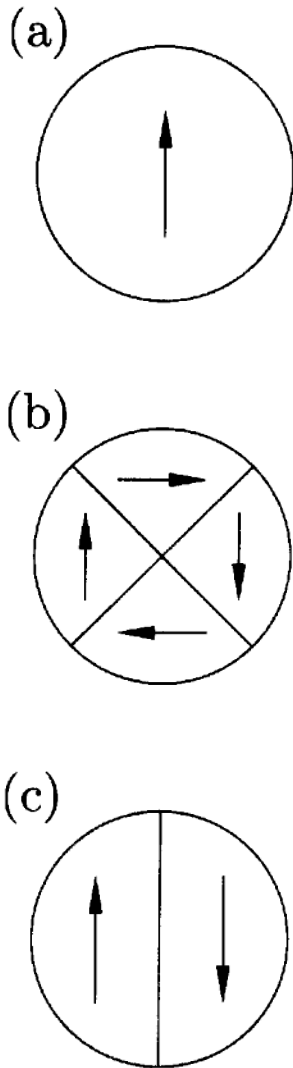


Fig. 6.26 Three possible magnetization configurations for a small spherical ferromagnetic particle.

The state in Fig. 6.26(c), appropriate for uniaxial anisotropy, is harder to consider because neither domain is ellipsoidal in shape and there is still some demagnetizing energy. Roughly, though, the demagnetizing energy of this configuration is half that of Fig. 6.26(a) since the domains are half the size. The domain wall cost is then $\pi r^2 \sigma_w^{180^\circ}$ where $\sigma_w^{180^\circ}$ is the energy of a 180° domain wall per unit area. Thus

$$E_{(c)} = \frac{1}{9} \mu_0 \pi M^2 r^3 + \pi r^2 \sigma_w^{180^\circ} \quad (6.53)$$

Hence the state in Fig. 6.26(a) becomes more favourable than the state in Fig. 6.26(c) when $E_{(a)} < E_{(c)}$, i.e. when

$$r < \frac{9\sigma_w^{180^\circ}}{\mu_0 M^2}. \quad (6.54)$$

If $\sigma_w^{180^\circ} \sim 10^{-2} \text{ Jm}^{-2}$ and $\mu_0 M \sim 1 \text{ T}$, the critical radius is $\sim 10^{-7} \text{ m}$. Note, if this value came out to be smaller than the domain wall width, then the calculation would be nonsensical.

6.7.8 The Stoner–Wohlfarth model

It is possible to calculate the magnetization curve for a single-domain particle, via the **Stoner–Wohlfarth model**. Because the particle is a single magnetic domain, there are no domain walls and one need only consider coherent domain rotation.

We consider a single-domain magnetic particle in a magnetic field H which is applied at an angle θ to the easy axis of its uniaxial anisotropy. If the magnetization of the particle then lies at an angle of ϕ to the magnetic field direction, the energy density of the system is

$$E = K \sin^2(\theta - \phi) - \mu_0 H M_s \cos \phi. \quad (6.55)$$

The energy can be minimized to find the direction of the magnetization at any given value of the applied magnetic field.

This is shown in Fig. 6.27 where the applied field is written in dimensionless units using the parameter h given by

$$h = \frac{\mu_0 M_s H}{2K} \quad (6.56)$$

for two choices of the direction of the applied magnetic field. Analytic solutions are possible for this model for $\theta = 0$ and $\theta = \pi/2$, but in the figure it is shown how they can be solved numerically. In each case the trajectory of the magnetization direction which lowers the energy is shown by a curve across an energy surface (energy as a function of ϕ and h) and the corresponding hysteresis loops are also shown. The hysteresis loops are shown for a larger choice of values of θ in Fig. 6.28, together with the calculated result for a polycrystalline average of directions, as would be appropriate for an array of single-domain particles (ignoring any interactions between the particles). This model demonstrates how the anisotropies present in a system can lead to hysteresis, even in a system in which there are no irreversible effects associated with domain-wall pinning.

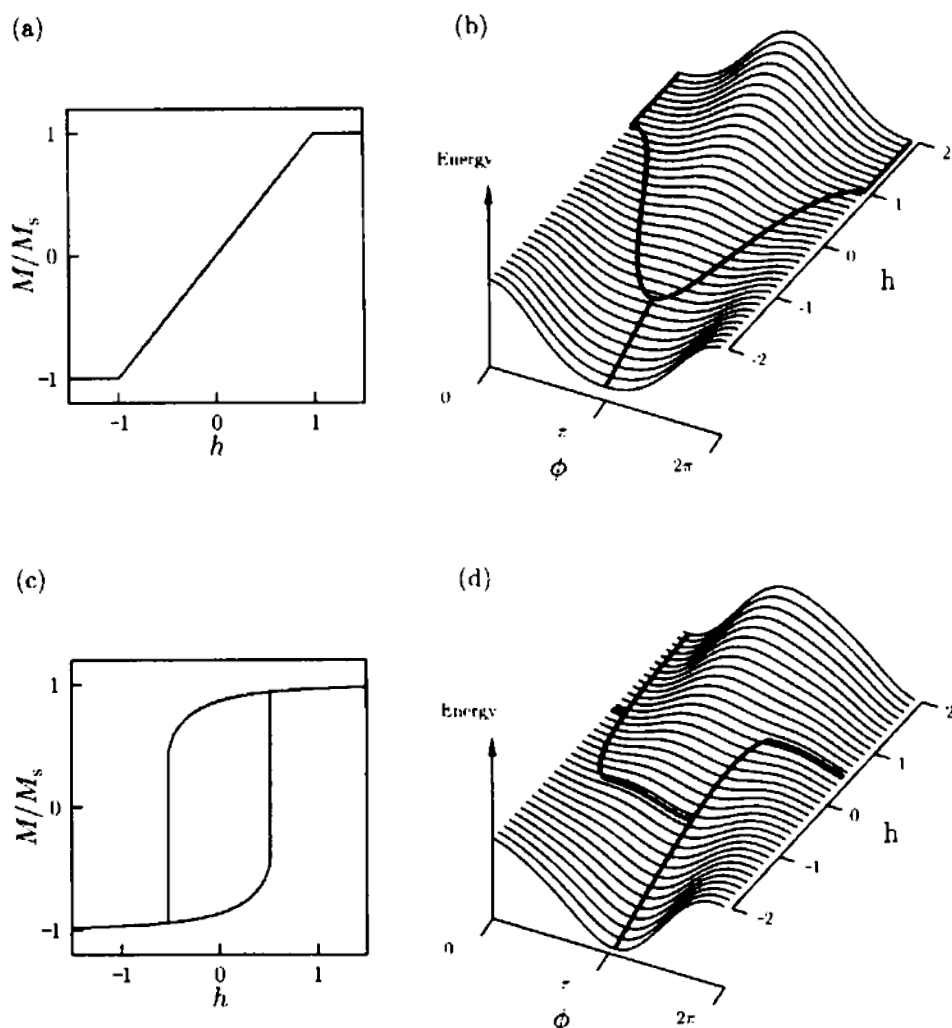


Fig. 6.27 The hysteresis loops in the Stoner-Wohlfarth model for (a) $\theta = 90^\circ$ and (c) $\theta = 30^\circ$. These can be obtained by finding the minimum energy points on the energy surfaces as a function of h and ϕ , which are shown for (b) $\theta = 90^\circ$ and (d) $\theta = 30^\circ$.

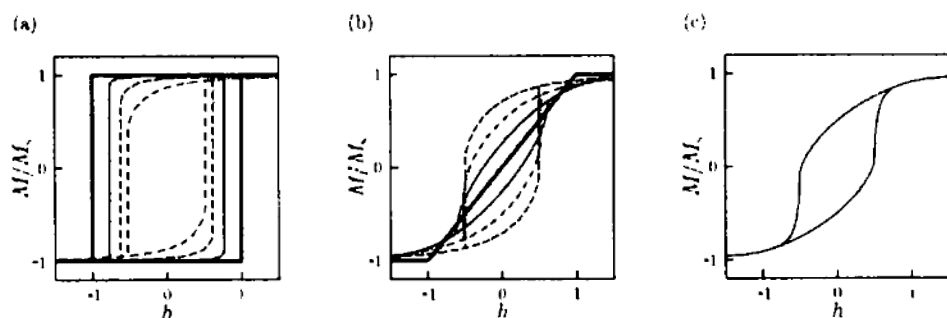


Fig. 6.28 The hysteresis loops in the Stoner-Wohlfarth model for (a) $\theta = 0^\circ$ (bold), 5° , 15° , 30° , (b) $\theta = 45^\circ$, 60° , 75° and 90° (bold). (c) A calculated hysteresis loop for a polycrystalline average.

6.7.9 Soft and hard materials

The energy dissipated (as heat) by a ferromagnet as it is taken around a circuit of its hysteresis loop is proportional to the area of that loop. If the area is small, the material is said to be magnetically **soft**. If the area is large, the material is said to be magnetically **hard**. Domain walls traverse a sample as the field is cycled and we can distinguish these two broad classes of ferromagnetic materials based on how easy it is for domains to move through a sample.

- (1) Soft magnetic materials are easy to magnetize. Soft magnets are used in transformer coils, generators and motors. In these applications the magnetization must be reversed many times a second and it is important that the energy dissipated in each cycle is minimized. Soft materials have broad domain walls (small anisotropy energy K) which are thus easy

to move. This leads to small coercive fields. Low magnetostriction is often desirable so that internal strains will not induce a local anisotropy energy. An example is permalloy (a commercial Ni/Fe alloy with an additional ingredient) which has a coercive field $B_c \sim 2 \times 10^{-7}$ T.

- (2) Hard magnetic materials are difficult to magnetize and thus difficult to demagnetize. Hard magnets are used as permanent magnets (e.g. in the back of loudspeakers, in motors, and of course on the front of your refrigerator!) and in magnetic tape recording (in powder form). In these applications, the magnetization needs to be preserved for as long as possible. The energy dissipated in a hysteresis loop cycle is as large as possible so that the magnetization will not occur spontaneously. Hard magnets have large hysteresis and narrow domain walls (large K) so that it is easy to have domain wall pinning. Large ion moments and large crystal fields are helpful for hard magnetic properties and suitable materials often involve rare earths, e.g. $\text{Nd}_2\text{Fe}_{14}\text{B}$ which has $T_C = 585$ K and coercive field $B_c = 1.2$ T.

An important application of hard magnetic materials is magnetic recording. The areal density of bits that can be recorded using magnetic materials has shot up dramatically over the last few years, reflecting a steady improvement in the understanding and fabrication of magnetic materials. The approximate figures are ~ 0.1 Mbits in^{-2} in 1960, ~ 10 Mbits in^{-2} in 1980 and ~ 25 Gbits in^{-2} in 2000 (1 bit in^{-2} is a bit of information, a '0' or '1', stored in a square-inch – this most modern of industries has yet to modernize its units!) As the areal density increases in the future, superparamagnetic fluctuations (i.e. thermal fluctuations of the magnetic particles, see Section 8.3) of the magnetic moments will become important.

Small magnetic particles of Fe_2O_3 were traditionally used on recording tapes and disks. Improvements in properties for high-fidelity audio recording and video tapes are based on high coercivity particles such as Fe_2O_3 doped with Co, CrO_2 , metal particles (usually Fe), and barium ferrite ($\text{BaO} \cdot 6\text{Fe}_2\text{O}_3$, see Section 5.3). Hard drives typically consist of a large rigid disk with a thin film deposited by vacuum deposition or sputtering, usually of a Co/Cr alloy. The hard disks rotate rapidly and a reading-head moves across the surface of the disk (it can hover a fraction of a micron above the surface) and is able to access any part of it.

Further reading

- The concept of broken symmetry is described in more detail in P. W. Anderson, *Basic notions of condensed matter physics*, Addison-Wesley 1984.
- The statistical mechanics of phase transitions is discussed in J. Yeomans, *Statistical mechanics of phase transitions*, OUP 1992.
- Also very helpful on critical phenomena is M. F. Collins, *Magnetic critical scattering*, OUP 1989.
- A very thorough description of phase transitions and broken symmetry phenomena, with particular emphasis on soft condensed matter, is P. M. Chaikin and T. C. Lubensky, *Principles of condensed matter physics*, CUP 1995.
- J. J. Binney, N. J. Dowrick, A. J. Fisher and M. E. J. Newman, *The theory of critical phenomena*, OUP 1992 contains an excellent account of renormalization group methods.
- Information on domains and magnetism may be found

in G. Bertoni, *Hysteresis in magnetism*, Academic Press 1998 and A. Hubert and R. Schäfer, *Magnetic domains*, Springer 1998.

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- S. Kaya, *Sci. Rep. Tohoku Univ.*, **17**, 639, 1157 (1928).
- Useful general references are D. Craik, *Magnetism: principles and applications*, Wiley 1995 and J. Crangle, *Solid state magnetism*, Edward Arnold 1991.
- T. G. Perring, G. Aeppli, S. M. Hayden, S. A. Carter, J. P. Remeika and S-W. Cheong, *Phys. Rev. Lett.*, **77**, 711 (1996).
- R. N. Sinclair and B. N. Brockhouse, *Phys. Rev.*, **120**, 1638 (1960).

Exercises

- (6.1) A one-dimensional ferromagnetic chain of N spins is described by the Ising Hamiltonian

$$\hat{H} = -2J \sum_{i=1}^{N-1} \hat{S}_i^z \hat{S}_{i+1}^z - 2J_N \hat{S}_N^z \hat{S}_1^z.$$

(The last term is used to give periodic boundary conditions.)

Show that the partition function Z of this system can be obtained by introducing new operators

$$\zeta_i = 4S_i^z S_{i+1}^z$$

which have eigenvalues $+1$ or -1 , and hence show that the partition function is $Z = (2 \cosh(J/2k_B T))^N$.

Using these results obtain an expression for the heat capacity per spin of the chain as $N \rightarrow \infty$. Deduce the low and high temperature behaviour of the heat capacity and sketch the heat capacity as a function of temperature. Discuss the result.

- (6.2) A uniaxial ferromagnet is described by the Hamiltonian

$$\hat{H} = - \sum_{ij} J_{ij} \hat{\mathbf{S}}_i \cdot \hat{\mathbf{S}}_j - \sum_{ij} K_{ij} \hat{S}_i^z \hat{S}_j^z. \quad (6.57)$$

(a) Show that the state with all spins fully aligned along the z axis is an eigenstate of the Hamiltonian.

(b) Obtain an expression for the spin wave spectrum as a function of wave vector $\mathbf{q}$.

(c) Simplify the expressions for the case where J_{ij} and K_{ij} are restricted to nearest neighbours, J_0 and K_0 , and the ferromagnet is (i) a one-dimensional chain, (ii) a two-dimensional square lattice and (iii) a three-dimensional body-centred cubic material.

- (6.3) Using the results of Exercise 6.2 for small wave vectors, deduce the temperature dependence at low temperatures

of the number of spin waves for the Heisenberg model ($K_0 = 0$) and the Ising model ($J_0 = 0$) for each of the structures (i), (ii) and (iii) of Exercise 6.2. Show that your results for the Ising model for case (i) agree with the results obtained in Exercise 6.1 and that there is no long range magnetic order above absolute zero for the Heisenberg model in one or two dimensions.

- (6.4) Apply spin-wave theory to an antiferromagnet and show that the dispersion relation is

$$\hbar\omega = 4|JS \sin(qa)|. \quad (6.58)$$

Hence show that for an antiferromagnet $\omega \propto |q|$ at small q , whereas for a ferromagnet $\omega \propto q^2$.

- (6.5) The Landau theory of a ferromagnet in a magnetic field H implies that the free energy is given by

$$F(M) = F_0 + a_0(T - T_C)M^2 + bM^4 - \mu_0 M H \quad (6.59)$$

where a_0 and b are positive constants. Show that this implies that

$$M^2 = u + v \frac{H}{M} \quad (6.60)$$

where u and v are constants that you should determine. By sketching M^2 against H/M for T just above T_C , just below T_C , and exactly at T_C , show how this method can be used to determine T_C . This idea is the basis of the **Arrott plot** which is a plot of M^2 against H/M .

- (6.6) Find analytic solutions to the Stoner–Wohlfarth model for $\theta = 0$ and $\theta = \pi/2$.

- (6.7) To illustrate the ideas behind the Mermin–Wagner–Berezinskii theorem, consider an excitation in two dimensions (see Fig. 6.29) in which the magnetization as a function of position $\mathbf{r} = (x, y)$ has constant magnitude $|\mathbf{M}|$ but varying direction specified by the vector $\mathbf{M}(\mathbf{r})$

given by

$$\mathbf{M}(\mathbf{r}) = \begin{cases} |\mathbf{M}|(\hat{\mathbf{y}} \cos(\pi r/R) - \hat{\mathbf{x}} \sin(\pi r/R)) & \text{if } r < R \\ -|\mathbf{M}|\hat{\mathbf{y}} & \text{if } r > R. \end{cases} \quad (6.61)$$

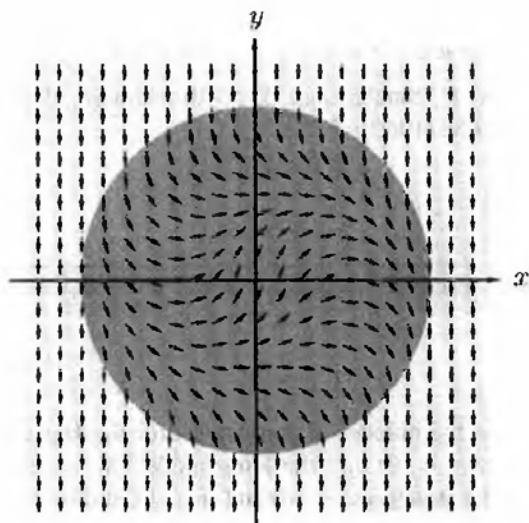


Fig. 6.29 An excitation in a 2D magnet.

Explain why the exchange energy cost due to this excitation is proportional to

$$\int \left[(\nabla M_x)^2 + (\nabla M_y)^2 \right] dx dy, \quad (6.62)$$

and show that it is a constant ($\pi^3 |\mathbf{M}|^2$) and therefore independent of R . This implies that very large scale fluctuations can occur with no additional energy cost compared with a small-scale fluctuation, thereby destroying any long range order. This problem is based on an example presented in an article by A. S. Arrott and B. Heinrich, *Journal of Magnetism and Magnetic Materials* **93**, 571 (1991).

(6.8) Data are written onto a magnetic recording disk with an areal density of 25 Gbits/in². By using some educated guesses, convert this areal density into other units: (a) bits per cross-sectional area of a hydrogen atom, (b) copies of the complete plays by William Shakespeare (i.e. their information content) per postage stamp. Compare this density of information storage to that of the human genome.

(6.9) Consider an isotropic cubic ferromagnet (lattice parameter a) in a magnetic field, with Hamiltonian

$$\hat{\mathcal{H}} = -J \sum_{\langle ij \rangle} \hat{\mathbf{S}}_i \cdot \hat{\mathbf{S}}_j + g\mu_B \sum_i \mathbf{B} \cdot \hat{\mathbf{S}}_i. \quad (6.63)$$

Show that it has a spin wave dispersion given by

$$\hbar\omega(\mathbf{q}) = g\mu_B B + 2JSq^2 a^2 \quad (6.64)$$

for $qa \ll 1$.

(6.10) Show that the Hamiltonian

$$\hat{\mathcal{H}} = - \sum_{\mathbf{n}, \delta} J(\delta) \hat{\mathbf{S}}_{\mathbf{n}} \cdot \hat{\mathbf{S}}_{\mathbf{n}+\delta} \quad (6.65)$$

has a spin-wave dispersion given by

$$\hbar\omega(\mathbf{q}) = \hbar\omega(0) + S \sum_{\delta} J(\delta) (\mathbf{q} \cdot \delta)^2 \quad (6.66)$$

for $|\mathbf{q} \cdot \delta_{\max}| \ll 1$ where $\delta_{\max}$ is the largest vector δ for which $J(\delta)$ gives any appreciable contribution.

(6.11) The state $|\mathbf{q}\rangle$ in eqn 6.23 represents a delocalized single flipped spin in a background of unflipped spins. In three dimensions it becomes

$$|\mathbf{q}\rangle = \frac{1}{\sqrt{N}} \sum_j e^{i\mathbf{q} \cdot \mathbf{R}_j} |\mathbf{j}\rangle. \quad (6.67)$$

The probability that the flipped spin is at site i is

$$P = |(q|i)|^2. \quad (6.68)$$

Show that $P = 1/N$ and hence that the lowered spin is distributed with equal probability.

The operator for the transverse spin correlation function is

$$\hat{S}_j^x \hat{S}_{j'}^x + \hat{S}_j^y \hat{S}_{j'}^y. \quad (6.69)$$

Show that

$$\langle \mathbf{q} | \hat{S}_j^x \hat{S}_{j'}^x + \hat{S}_j^y \hat{S}_{j'}^y | \mathbf{q} \rangle = \frac{2S}{N} \cos(\mathbf{q} \cdot (\mathbf{R}_j - \mathbf{R}_{j'})) \quad (6.70)$$

(for $j \neq j'$) and interpret this with reference to Fig. 6.13.

(6.12) Our calculation for the domain wall width of a Bloch wall made the assumption that the angle between adjacent spins in the wall was π/N everywhere throughout the N spins of the wall. This was clearly an approximation! Here is an improved calculation.

Let the spin as a function of z , a coordinate running perpendicular to the plane of the wall, be described by an angle $\theta(z)$. We assume that $\theta(z) \rightarrow 0$ as $z \rightarrow -\infty$ and $\theta(z) \rightarrow \pi$ as $z \rightarrow +\infty$, corresponding to a 180° wall. The energy is given by

$$E = \int_{-\infty}^{\infty} \left[A \left(\frac{\partial \theta}{\partial z} \right)^2 + f(\theta) \right] dz, \quad (6.71)$$

where A is the continuum exchange constant and $f(\theta)$ is a function which describes the anisotropy. Show that the energy is minimized when

$$\frac{\partial f}{\partial \theta} - 2A \frac{\partial^2 f}{\partial z^2} = 0. \quad (6.72)$$

Multiply this equation by $\partial\theta/\partial z$ and integrate with respect to z to show that

$$f(\theta) = A \left(\frac{\partial f}{\partial z} \right)^2. \quad (6.73)$$

Now suppose that $f(\theta)$ is given by

$$f(\theta) = K \sin^2 \theta, \quad (6.74)$$

where K is an anisotropy constant. Show that this implies that

$$\cos \theta(z) = -\tanh \left(\frac{\pi z}{\delta} \right), \quad (6.75)$$

where $\delta = \pi \sqrt{A/K}$. Sketch $\theta(z)$ and argue that δ is a suitable measure of the domain wall width. Show further that

$$E = 4\sqrt{AK} \quad (6.76)$$

in this case.

- (6.13) Consider the closure domain structure in a ferromagnetic film as shown in Fig. 6.30. The easy axis is vertical and the hard axis is horizontal. The energy of the long 180° walls per unit area is σ_w and the anisotropy energy density is K . Ignoring the contribution from the 90° walls

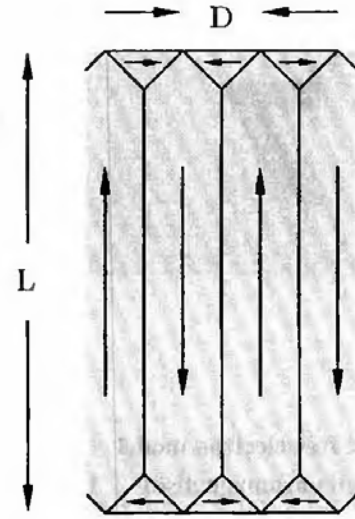


Fig. 6.30 A closure domain.

(i.e. assume $L \gg D$), show that

$$D \approx \sqrt{\frac{2\sigma_w L}{K}}. \quad (6.77)$$

Assuming that $\sigma_w = 2 \times 10^{-3} \text{ Jm}^{-2}$ and that $K = 4 \times 10^4 \text{ Jm}^{-3}$ and $L = 4 \text{ mm}$, estimate D .